



REALTEK

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RTL8367-VB-CG

SINGLE-CHIP 5-PORT 10/100/1000 SWITCH CONTROLLER

DATASHEET

(CONFIDENTIAL: Development Partners Only)

Rev. Pre-0.9

10 Jan. 2011

Track ID: xxxx-xxxx-xx



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USING THIS DOCUMENT

This document is intended for the hardware and software engineer’s general information on the Realtek RTL8367-VB IC.

Though every effort has been made to ensure that this document is current and accurate, more information may have become available subsequent to the production of this guide.

REVISION HISTORY

Revision	Release Date	Summary
Pre-0.9	2011/1/10	Preliminary Release.

Table of Contents

1. GENERAL DESCRIPTION	1
2. FEATURES	3
3. SYSTEM APPLICATIONS	4
4. APPLICATION EXAMPLE	5
4.1. 5-PORT 1000BASE-T SWITCH	5
5. BLOCK DIAGRAM	6
6. PIN ASSIGNMENTS	7
6.1. PACKAGE IDENTIFICATION	8
6.2. PIN ASSIGNMENT TABLE	8
7. PIN DESCRIPTIONS	10
7.1. MEDIA DEPENDENT INTERFACE PINS	10
7.2. LED PINS	11
7.3. CONFIGURATION STRAPPING PINS	12
7.4. MISCELLANEOUS PINS	14
7.5. TEST PINS	15
7.6. POWER AND GND PINS	15
7.7. NC (NO CONNECTION) PINS	15
8. PHYSICAL LAYER FUNCTIONAL OVERVIEW	16
8.1. MDI INTERFACE	16
8.2. 1000BASE-T TRANSMIT FUNCTION	16
8.3. 1000BASE-T RECEIVE FUNCTION	16
8.4. 100BASE-TX TRANSMIT FUNCTION	16
8.5. 100BASE-TX RECEIVE FUNCTION	17
8.6. 10BASE-T TRANSMIT FUNCTION	17
8.7. 10BASE-T RECEIVE FUNCTION	17
8.8. AUTO-NEGOTIATION FOR UTP	17
8.9. Crossover DETECTION AND AUTO CORRECTION	18
8.10. POLARITY CORRECTION	18
9. GENERAL FUNCTION DESCRIPTION	19
9.1. RESET	19
9.1.1. <i>Hardware Reset</i>	19
9.1.2. <i>Software Reset</i>	19
9.2. IEEE 802.3x FULL DUPLEX FLOW CONTROL	19
9.3. HALF DUPLEX FLOW CONTROL	20
9.3.1. <i>Back-Pressure Mode</i>	20
9.4. SEARCH AND LEARNING	21
9.5. SVL AND IVL/SVL	21
9.6. ILLEGAL FRAME FILTERING	21
9.7. IEEE 802.3 RESERVED GROUP ADDRESSES FILTERING CONTROL	22
9.8. BROADCAST/MULTICAST/UNKNOWN DA STORM CONTROL	22
9.9. PORT SECURITY FUNCTION	23
9.10. MIB COUNTERS	24
9.11. PORT MIRRORING	24
9.12. VLAN FUNCTION	24
9.12.1. <i>Port-Based VLAN</i>	24

9.12.2.	IEEE 802.1Q Tag-Based VLAN.....	25
9.12.3.	Protocol-Based VLAN.....	26
9.12.4.	Port VID.....	26
9.13.	QoS FUNCTION.....	27
9.13.1.	Input Bandwidth Control.....	27
9.13.2.	Priority Assignment.....	27
9.13.3.	Priority Queue Scheduling.....	28
9.13.4.	IEEE 802.1p/Q and DSCP Remarking.....	28
9.13.5.	ACL-Based Priority.....	29
9.14.	IGMP & MLD SNOOPING FUNCTION.....	29
9.15.	IEEE 802.1x FUNCTION.....	29
9.15.1.	Port-Based Access Control.....	29
9.15.2.	Authorized Port-Based Access Control.....	30
9.15.3.	Port-Based Access Control Direction.....	30
9.15.4.	MAC-Based Access Control.....	30
9.15.5.	MAC-Based Access Control Direction.....	30
9.15.6.	Optional Unauthorized Behavior.....	30
9.15.7.	Guest VLAN.....	30
9.16.	IEEE 802.1D FUNCTION.....	31
9.17.	EMBEDDED 8051.....	31
9.18.	REALTEK CABLE TEST (RTCT).....	31
9.19.	LED INDICATOR.....	32
9.20.	GREEN ETHERNET.....	33
9.20.1.	Link-On and Cable Length Power Saving.....	33
9.20.2.	Link-Down Power Saving.....	33
9.21.	IEEE 802.3AZ ENERGY EFFICIENT ETHERNET (EEE) FUNCTION.....	34
10.	INTERFACE DESCRIPTIONS.....	35
10.1.	EEPROM SMI HOST TO EEPROM.....	35
10.2.	EEPROM SMI SLAVE FOR EXTERNAL CPU.....	36
11.	REGISTER DESCRIPTIONS.....	37
11.1.	PCS REGISTER (PHY 0~4).....	37
11.2.	REGISTER 0: CONTROL.....	38
11.3.	REGISTER 1: STATUS.....	39
11.4.	REGISTER 2: PHY IDENTIFIER 1.....	40
11.5.	REGISTER 3: PHY IDENTIFIER 2.....	40
11.6.	REGISTER 4: AUTO-NEGOTIATION ADVERTISEMENT.....	40
11.7.	REGISTER 5: AUTO-NEGOTIATION LINK PARTNER ABILITY.....	41
11.8.	REGISTER 6: AUTO-NEGOTIATION EXPANSION.....	42
11.9.	REGISTER 7: AUTO-NEGOTIATION PAGE TRANSMIT REGISTER.....	42
11.10.	REGISTER 8: AUTO-NEGOTIATION LINK PARTNER NEXT PAGE REGISTER.....	43
11.11.	REGISTER 9: 1000BASE-T CONTROL REGISTER.....	43
11.12.	REGISTER 10: 1000BASE-T STATUS REGISTER.....	44
11.13.	REGISTER 15: EXTENDED STATUS.....	44
12.	ELECTRICAL CHARACTERISTICS.....	45
12.1.	ABSOLUTE MAXIMUM RATINGS.....	45
12.2.	RECOMMENDED OPERATING RANGE.....	45
12.3.	THERMAL CHARACTERISTICS.....	46
12.3.1.	QFP-128.....	46
12.4.	DC CHARACTERISTICS.....	48
12.5.	AC CHARACTERISTICS.....	48
12.5.1.	EEPROM SMI Host Mode Timing Characteristics.....	48
12.5.2.	EEPROM SMI Slave Mode Timing Characteristics.....	50

12.5.3.	MDIO Slave Mode Timing Characteristics	50
12.6.	POWER AND RESET CHARACTERISTICS	52
13.	MECHANICAL DIMENSIONS.....	53
13.1.	MECHANICAL DIMENSIONS NOTES	54
14.	ORDERING INFORMATION.....	55

List of Tables

TABLE 1. PIN ASSIGNMENT TABLE	8
TABLE 2. MEDIA DEPENDENT INTERFACE PINS	10
TABLE 3. LED PINS	11
TABLE 4. CONFIGURATION STRAPPING PINS	12
TABLE 5. MISCELLANEOUS PINS	14
TABLE 6. TEST PINS	15
TABLE 7. POWER AND GND PINS	15
TABLE 8. NC (NO CONNECTION) PIN	15
TABLE 9. MEDIA DEPENDENT INTERFACE PIN MAPPING	18
TABLE 10. RESERVED MULTICAST ADDRESS CONFIGURATION TABLE	22
TABLE 11. LED DEFINITIONS	32
TABLE 12. PCS REGISTER (PHY 0~4)	37
TABLE 13. REGISTER 0: CONTROL	38
TABLE 14. REGISTER 1: STATUS	39
TABLE 15. REGISTER 2: PHY IDENTIFIER 1	40
TABLE 16. REGISTER 3: PHY IDENTIFIER 2	40
TABLE 17. REGISTER 4: AUTO-NEGOTIATION ADVERTISEMENT	40
TABLE 18. REGISTER 5: AUTO-NEGOTIATION LINK PARTNER ABILITY	41
TABLE 19. REGISTER 6: AUTO-NEGOTIATION EXPANSION	42
TABLE 20. REGISTER 7: AUTO-NEGOTIATION PAGE TRANSMIT REGISTER	42
TABLE 21. REGISTER 8: AUTO-NEGOTIATION LINK PARTNER NEXT PAGE REGISTER	43
TABLE 22. REGISTER 9: 1000BASE-T CONTROL REGISTER	43
TABLE 23. REGISTER 10: 1000BASE-T STATUS REGISTER	44
TABLE 24. REGISTER 15: EXTENDED STATUS	44
TABLE 25. ABSOLUTE MAXIMUM RATINGS	45
TABLE 26. RECOMMENDED OPERATING RANGE	45
TABLE 27. ASSEMBLY DESCRIPTION	46
TABLE 28. MATERIAL PROPERTIES	46
TABLE 29. SIMULATION CONDITIONS	46
TABLE 30. THERMAL PERFORMANCE OF QFP-128 ON PCB UNDER STILL AIR CONVECTION	47
TABLE 31. THERMAL PERFORMANCE OF QFP-128 ON PCB UNDER FORCED CONVECTION	47
TABLE 32. DC CHARACTERISTICS	48
TABLE 33. EEPROM SMI HOST MODE TIMING CHARACTERISTICS	49
TABLE 34. EEPROM SMI SLAVE MODE TIMING CHARACTERISTICS	50
TABLE 35. MDIO TIMING CHARACTERISTICS AND REQUIREMENTS	51
TABLE 36. POWER AND RESET CHARACTERISTICS	52
TABLE 37. ORDERING INFORMATION	55

List of Figures

FIGURE 1. 5-PORT 1000BASE-T SWITCH	5
FIGURE 2. BLOCK DIAGRAM.....	6
FIGURE 3. PIN ASSIGNMENTS	7
FIGURE 4. CONCEPTUAL EXAMPLE OF POLARITY CORRECTION	18
FIGURE 5. PROTOCOL-BASED VLAN FRAME FORMAT AND FLOW CHART.....	26
FIGURE 6. MAX-MIN SCHEDULING DIAGRAM	28
FIGURE 7. PULL-UP AND PULL-DOWN OF LED PINS FOR SINGLE-COLOR LED.....	33
FIGURE 8. PULL-UP AND PULL-DOWN OF LED PINS FOR BI-COLOR LED.....	33
FIGURE 9. SMI START AND STOP COMMAND	35
FIGURE 10. EEPROM SMI HOST TO EEPROM.....	35
FIGURE 11. EEPROM SMI HOST MODE FRAME.....	35
FIGURE 12. EEPROM SMI WRITE COMMAND FOR SLAVE MODE	36
FIGURE 13. EEPROM SMI READ COMMAND FOR SLAVE MODE.....	36
FIGURE 14. EEPROM SMI HOST MODE TIMING CHARACTERISTICS.....	48
FIGURE 15. SCK/SDA POWER ON TIMING	49
FIGURE 16. EEPROM AUTO-LOAD TIMING.....	49
FIGURE 17. EEPROM SMI SLAVE MODE TIMING CHARACTERISTICS	50
FIGURE 18. MDIO SOURCED BY MASTER (RTL8367 LINK PARTNER CPU).....	51
FIGURE 19. MDIO SOURCED BY SLAVE (RTL8367)	51
FIGURE 20. POWER AND RESET CHARACTERISTICS.....	52

1. General Description

The RTL8367 is a QFP128, high-performance 5-port Gigabit Ethernet switch with an integrated low-power 5-port Giga-PHY that supports 1000Base-T, 100Base-TX, and 10Base-T.

The RTL8367 integrates all the functions of a high-speed switch system; including SRAM for packet buffering, non-blocking switch fabric, and internal register management into a single CMOS device. Only a 25MHz crystal is required; an optional EEPROM is offered for internal register configuration.

The embedded packet storage SRAM in the RTL8367 features superior memory management technology to efficiently utilize memory space. The RTL8367 integrates a 2K-entry look-up table with a 4-way XOR Hashing algorithm for address searching and learning. The table provides read/write access from the EEPROM Serial Management Interface (SMI), and each of the entries can be configured as a static entry. The entry aging time is between 200 and 400 seconds. Eight Filtering Databases are used to provide Independent VLAN Learning and Shared VLAN Learning (IVL/SVL) functions.

The RTL8367 supports Port VID (PVID) for each port to insert a PVID in the VLAN tag on egress. When using this function, VID information carried in the VLAN tag will be changed to PVID.

The RTL8367 supports standard 802.3x flow control frames for full duplex, and optional backpressure for half duplex. It determines when to invoke the flow control mechanism by checking the availability of system resources, including the packet buffers and transmitting queues. The RTL8367 supports broadcast/multicast output dropping, and will forward broadcast/multicast packets to non-blocked ports only. For IP multicast applications, the RTL8367 supports IPv4 IGMPv1/v2/v3 and IPv6 MLDv1/v2 snooping.

In order to support flexible traffic classification, the RTL8367 supports 64-entry ACL rule check and multiple actions options. Each port can optionally enable or disable the ACL rule check function. The ACL rule key can be based on packet physical port, Layer2, Layer3, and Layer4 information. When an ACL rule matches, the action taken is configurable to Drop/Permit/Redirect/ Mirror, change priority value in 802.1p/Q tag, and rate policing. The rate policing mechanism supports from 8Kbps to 1Gbps (in 8Kbps steps).

In Bridge operation the RTL8367 supports 16 sets of port configurations: disable, block, learning, and forwarding for Spanning Tree Protocol and Multiple Spanning Tree Protocol. To meet security and management application requirements, the RTL8367 supports IEEE 802.1x Port-based/MAC-based Access Control. For those ports that do not pass IEEE 802.1x authentication, the RTL8367 provides a Port-based/MAC-based Guest VLAN function for them to access limited network resources. A 1-set Port Mirroring function is configured to mirror traffic (RX, TX, or both) appearing on one of the switch's ports. Support is provided on each port for multiple RFC MIB Counters, for easy debug and diagnostics.

To improve real-time or multimedia networking applications, the RTL8367 supports eight priority assignments for each received packet. These are based on (1) Port-based priority; (2) 802.1p/Q VLAN tag priority; (3) DSCP field in IPv4/IPv6 header; and (4) ACL-assigned priority. Each output port supports a weighted ratio of eight priority queues to fit bandwidth requirements in different applications. The input bandwidth control function helps limit per-port traffic utilization. There is one leaky bucket for average packet rate control for each queue of all ports. Queue scheduling algorithm can use Strict Priority (SP) or Weighted Fair Queue (WFQ) or mixed.

The RTL8367 provides a 4K-entry VLAN table for 802.1Q port-based, tag-based, and protocol-based VLAN operation to separate logical connectivity from physical connectivity. The RTL8367 supports four Protocol-based VLAN configurations that can optionally select EtherType, LLC, and RFC1042 as the search key. Each port may be set to any topology via EEPROM upon reset, or EEPROM SMI Slave after reset.

In router applications, the router may want to know the input port of the incoming packet. The RTL8367 supports an option to insert a VLAN tag with VID=Port VID (PVID) on each egress port. The RTL8367 also provides an option to admit VLAN tagged packet with a specific PVID only. If this function is enabled, the RTL8367 will drop all non-tagged packets and packets with an incorrect PVID.

2. Features

- Single-chip 5-port gigabit non-blocking switch architecture
- Embedded 5-port 10/100/1000Base-T PHY
- Each port supports full duplex 10/100/1000M connectivity (half duplex only supported in 10/100M mode)
- Full-duplex and half-duplex operation with IEEE 802.3x flow control and backpressure
- Supports 9216-byte jumbo packet length forwarding at wire speed
- Supports Realtek Cable Test (RTCT) function
- Supports 64-entry ACL Rules
 - ◆ Search keys support physical port, Layer2, Layer3, and Layer4 information
 - ◆ Actions support mirror, redirect, dropping, priority adjustment, traffic policing, CVLAN decision, SVLAN assignment, GPIO control, interrupt, and logging counter
 - ◆ Supports 5 types of user defined ACL rule format for 64 ACL rules
 - ◆ Optional per-port enable/disable of ACL function
 - ◆ Optional setting of per-port action to take when ACL mismatch
- Supports IEEE 802.1Q VLAN
 - ◆ Supports 4K VLANs and 32 Extra Enhanced VLANs
 - ◆ Supports Un-tag definition in each VLAN
 - ◆ Supports VLAN policing and VLAN forwarding decision
 - ◆ Supports Port-based, Tag-based, and Protocol-based VLAN
 - ◆ Up to 4 Protocol-based VLAN entries
 - ◆ Supports per-port and per-VLAN egress VLAN tagging and un-tagging
- Supports IVL, SVL, and IVL/SVL
 - ◆ Supports 2K-entry MAC address table with 4-way hash algorithm
 - ◆ Up to 2K L2/L3 Filtering Database
 - ◆ Per-port MAC leaning limitation
- Supports Spanning Tree port behavior configuration
 - ◆ IEEE 802.1w Rapid Spanning Tree
 - ◆ IEEE 802.1s Multiple Spanning Tree with up to 16 Spanning Tree instances
- Supports IEEE 802.1x Access Control Protocol
 - ◆ Port-Based Access Control
 - ◆ MAC-Based Access Control
 - ◆ Guest VLAN
- Support Auto-doS
- Support H/W IGMP/MLD Snooping
 - ◆ IGMPv1/v2/3 and MLD v1/v2
 - ◆ Support Fast Leave
 - ◆ Static router port configuration
 - ◆ Dynamic router port learning and aging
- Supports Quality of Service (QoS)
 - ◆ Supports per port Input Bandwidth Control
 - ◆ Traffic classification based on IEEE 802.1p/Q priority definition, physical

- Port, IP DSCP field, ACL definition, VLAN based priority, MAC based priority and SVLAN based priority
 - ◆ Eight Priority Queues per port
 - ◆ Per queue flow control
 - ◆ Min-Max Scheduling
 - ◆ Strict Priority and Weighted Fair Queue (WFQ) to provide minimum bandwidth
 - ◆ One leaky bucket to constrain the average packet rate of each queue
- Supports rate limiting (64 shared meters, with 8kpbs granulation)
- Supports RFC MIB Counter
 - ◆ MIB-II (RFC 1213)
 - ◆ Ethernet-Like MIB (RFC 3635)
 - ◆ Interface Group MIB (RFC 2863)
 - ◆ RMON (RFC 2819)
 - ◆ Bridge MIB (RFC 1493)
 - ◆ Bridge MIB Extension (RFC 2674)
- Supports Stacking VLAN and Port Isolation with 8 Enhanced Filtering Databases
- Supports IEEE 802.1ad Stacking VLAN
 - ◆ Supports 64 SVLANs
 - ◆ Supports 32 L2/IPv4 Multicast mappings to SVLAN
 - ◆ Supports MAC-based 1:N VLAN
- Supports 2 IEEE 802.3ad Link aggregation port groups
- Supports Port Mirror function for one source port to multiple destination ports
- Supports OAM and IEEE LLDP (Energy Efficient Ethernet Link Layer Discovery Protocol)
- Supports Loop Detection
- Security Filtering
 - ◆ Disable learning for each port
 - ◆ Disable learning-table aging for each port
 - ◆ Drop unknown DA for each port
- Broadcast/Multicast/Unknown DA storm control protects system from attack by hackers
- Supports Realtek Green Ethernet features
 - ◆ Link-On Cable Length Power Saving
 - ◆ Link-Down Power Saving
- Each port supports 3 LED outputs
- Supports EEPROM SMI Slave interface to access configuration register
- Supports 32K-byte EEPROM space for configuration
- Integrated 8051 microprocessor.
- 25MHz crystal or 3.3V OSC input
- QFP 128-pin package

3. System Applications

- 5-Port 1000Base-T Switch

4. Application Example

4.1. 5-Port 1000Base-T Switch

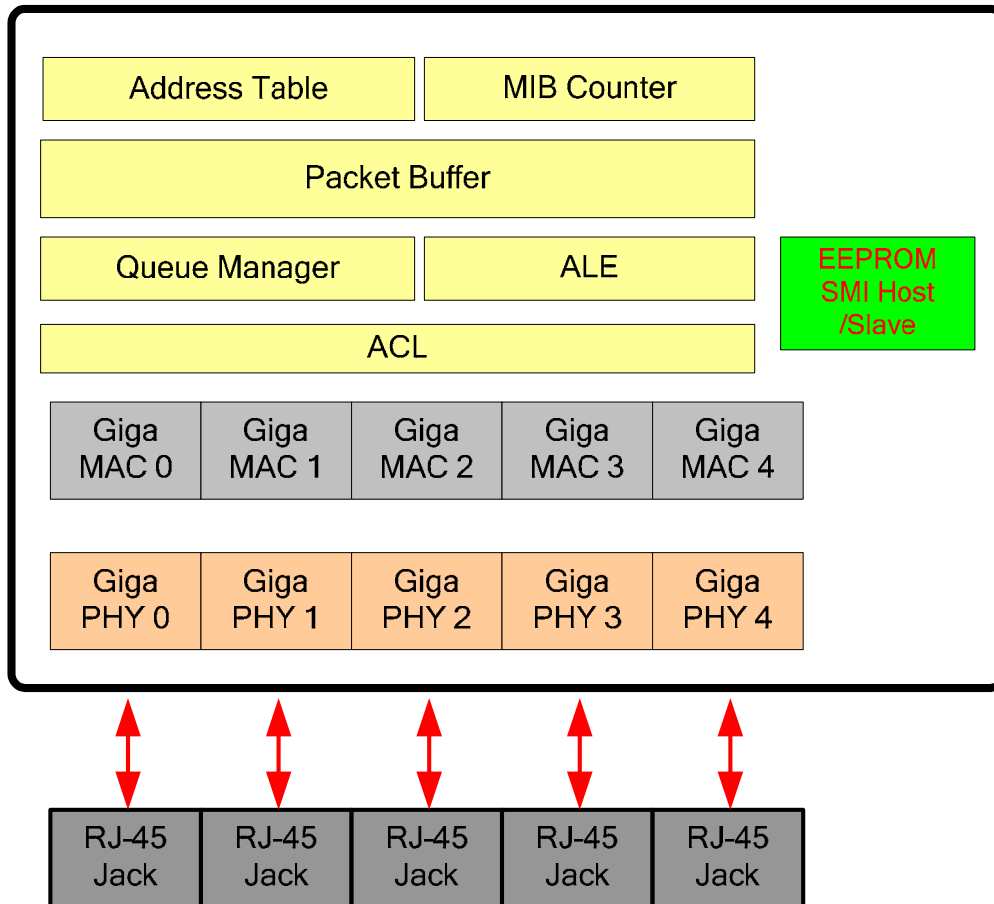


Figure 1. 5-Port 1000Base-T Switch

5. Block Diagram

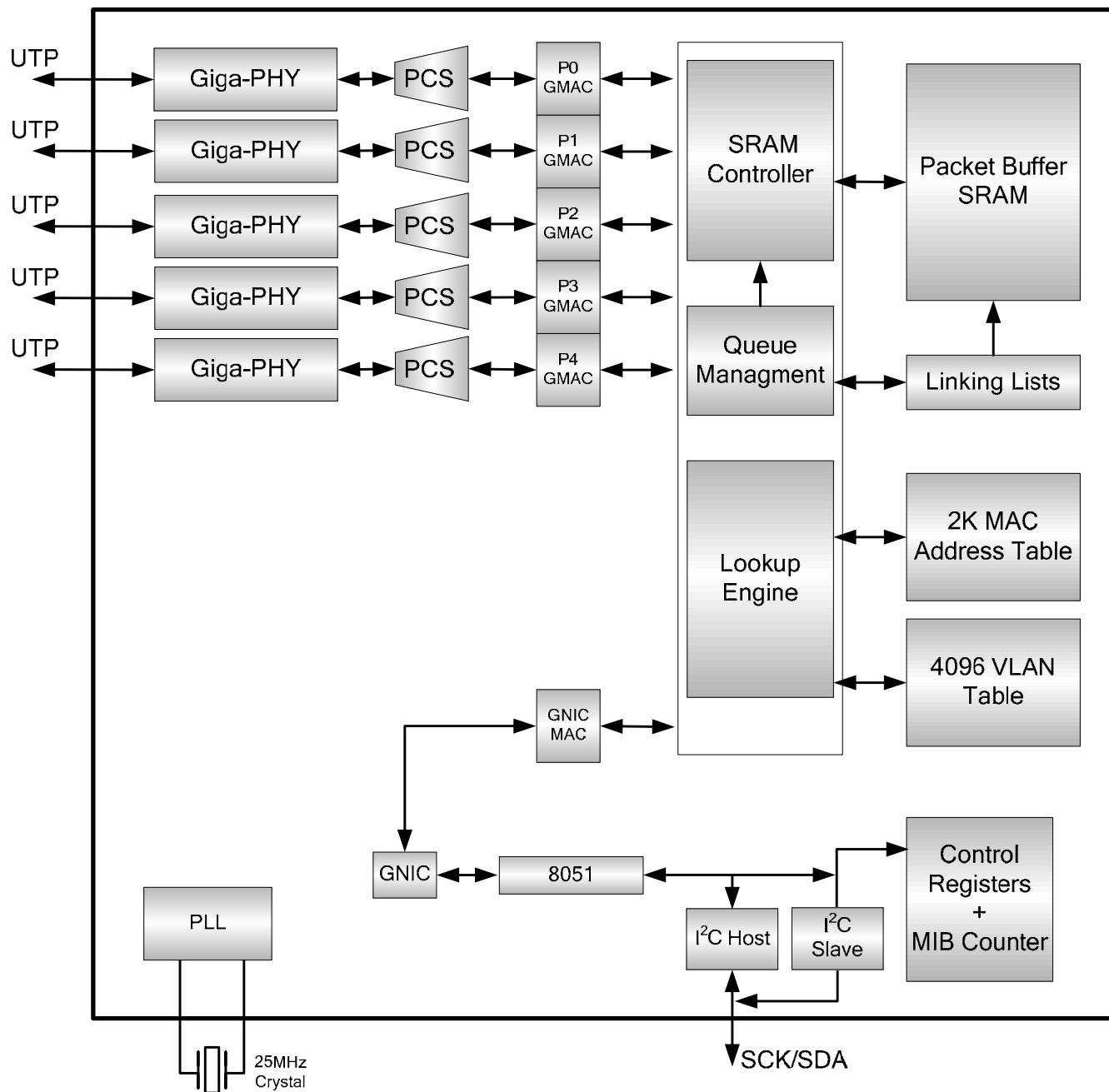


Figure 2. Block Diagram

6. Pin Assignments

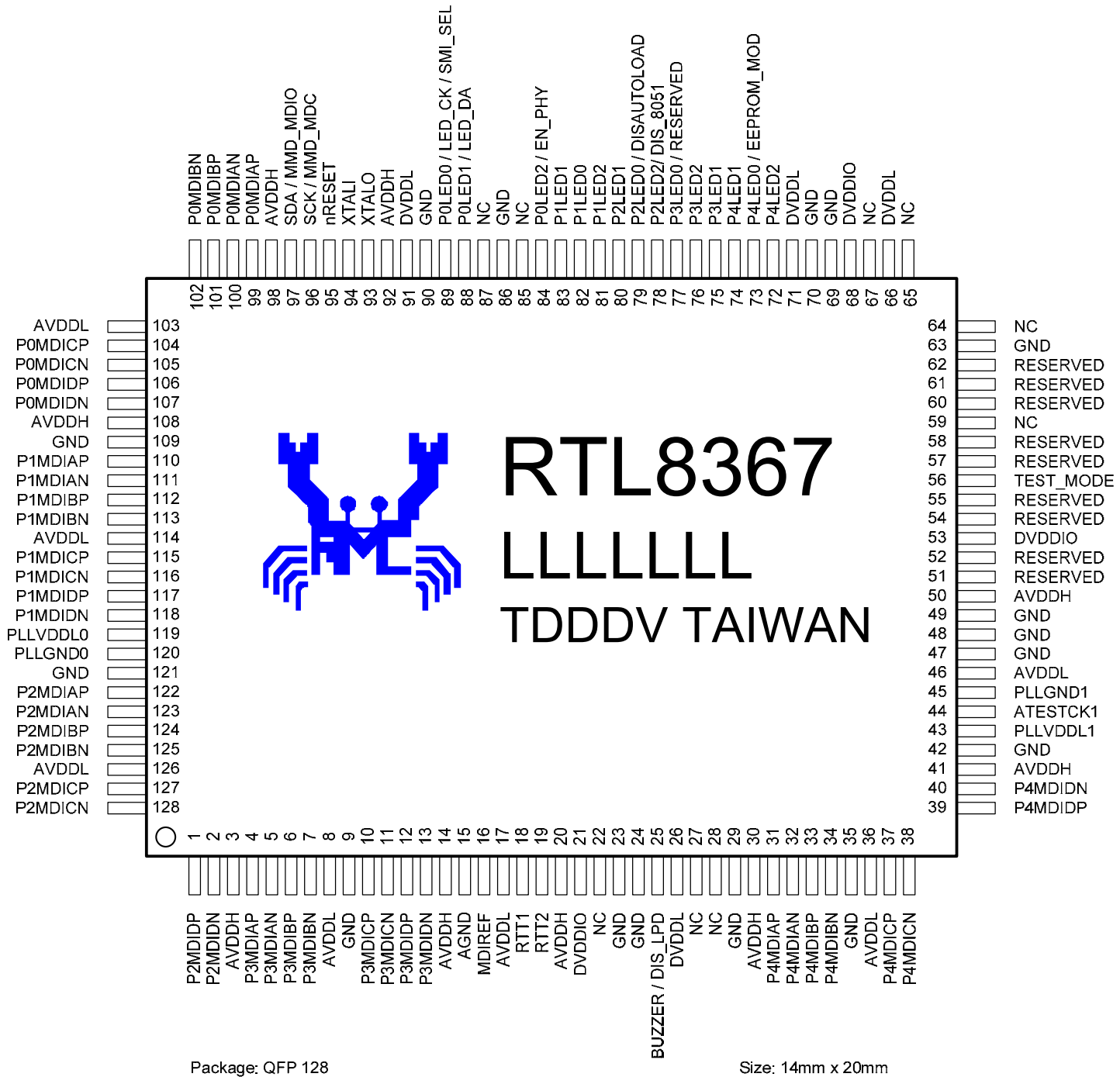


Figure 3. Pin Assignments

6.1. Package Identification

The RTL8367 package identification is showed in Figure 3.

Lot number is indicated in the location marked ‘L’.

Green package is indicated by a ‘G’ in the location marked ‘T’.

Date code is indicated in the location marked ‘D’.

Version number is indicated in the location marked ‘V’

6.2. Pin Assignment Table

Upon Reset: Defined as a short time after the end of a hardware reset.

After Reset: Defined as the time after the specified ‘Upon Reset’ time.

I: Input Pin	AI: Analog Input Pin
O: Output Pin	AO: Analog Output Pin
I/O: Bi-Direction Input/Output Pin	AI/O: Analog Bi-Direction Input/Output Pin
P: Digital Power Pin	AP: Analog Power Pin
G: Digital Ground Pin	AG: Analog Ground Pin
I _{PU} : Input Pin With Pull-Up Resistor; (Typical Value = 75K Ohm)	O _{PU} : Output Pin With Pull-Up Resistor; (Typical Value = 75K Ohm)
I _S : Input Pin With Schmitt Trigger	

Table 1. Pin Assignment Table

Name	Pin No.	Type	Name	Pin No.	Type
P2MDIDP	1	AI/O	P3MDIDP	12	AI/O
P2MDIDN	2	AI/O	P3MDIDN	13	AI/O
AVDDH	3	AP	AVDDH	14	AP
P3MDIAP	4	AI/O	AGND	15	AG
P3MDIAN	5	AI/O	MDIREF	16	AO
P3MDIBP	6	AI/O	AVDDL	17	AP
P3MDIBN	7	AI/O	RTT1	18	AO
AVDDL	8	AP	RTT2	19	AO
GND	9	G	AVDDH	20	AP
P3MDICP	10	AI/O	DVDDIO	21	P
P3MDICN	11	AI/O	NC	22	-

Name	Pin No.	Type
GND	23	G
GND	24	G
BUZZER/DIS_LPD	25	I/O _{PU}
DVDDL	26	P
NC	27	-
NC	28	-
GND	29	G
AVDDH	30	AP
P4MDIAP	31	AI/O
P4MDIAN	32	AI/O
P4MDIBP	33	AI/O
P4MDIBN	34	AI/O
GND	35	G
AVDDL	36	AP
P4MDICP	37	AI/O
P4MDICN	38	AI/O
P4MDIDP	39	AI/O
P4MDIDN	40	AI/O
AVDDH	41	AP
GND	42	G
PLLVDL1	43	AP
ATESTCK1	44	AO
PLLGND1	45	AG
AVDDL	46	AP
GND	47	G
GND	48	G
GND	49	G
AVDDH	50	AP
RESERVED	51	AO
RESERVED	52	AO
DVDDIO	53	P
RESERVED	54	I/O
RESERVED	55	I/O
TEST_MODE	56	I _{PD}
RESERVED	57	O
RESERVED	58	I/O
NC	59	-
RESERVED	60	I/O
RESERVED	61	O
RESERVED	62	I/O
GND	63	G
NC	64	-
NC	65	-
DVDDL	66	P

Name	Pin No.	Type
NC	67	-
DVDDIO	68	P
GND	69	G
GND	70	G
DVDDL	71	P
P4LED2	72	I/O _{PU}
P4LED0/EEPROM_MOD	73	I/O _{PU}
P4LED1	74	I/O _{PU}
P3LED1	75	I/O _{PU}
P3LED2	76	I/O _{PU}
P3LED0/RESERVED	77	I/O _{PU}
P2LED2/DIS_8051	78	I/O _{PU}
P2LED0/DISAUTOLOAD	79	I/O _{PU}
P2LED1	80	I/O _{PU}
P1LED2	81	I/O _{PU}
P1LED0	82	I/O _{PU}
P1LED1	83	I/O _{PU}
P0LED2/EN_PHY	84	I/O _{PU}
NC	85	-
GND	86	G
NC	87	-
P0LED1/LED_DA	88	I/O _{PU}
P0LED0/LED_CK/SMI_SEL	89	I/O _{PU}
GND	90	G
DVDDL	91	P
AVDDH	92	AP
XTALO	93	AO
XTALI	94	AI
nRESET	95	I _S
SCK/MMD_MDC	96	I/O _{PU}
SDA/MMD_MDIO	97	I/O _{PU}
AVDDH	98	P
P0MDIAP	99	AI/O
P0MDIAN	100	AI/O
P0MDIBP	101	AI/O
P0MDIBN	102	AI/O
AVDDL	103	AP
P0MDICP	104	AI/O
P0MDICN	105	AI/O
P0MDIDP	106	AI/O
P0MDIDN	107	AI/O
AVDDH	108	AP
GND	109	G
P1MDIAP	110	AI/O

Name	Pin No.	Type
P1MDIAN	111	AI/O
P1MDIBP	112	AI/O
P1MDIBN	113	AI/O
AVDDL	114	AP
P1MDICP	115	AI/O
P1MDICN	116	AI/O
P1MDIDP	117	AI/O
P1MDIDN	118	AI/O
PLLVDL0	119	AP

Name	Pin No.	Type
PLLGND0	120	AG
GND	121	G
P2MDIAP	122	AI/O
P2MDIAN	123	AI/O
P2MDIBP	124	AI/O
P2MDIBN	125	AI/O
AVDDL	126	AP
P2MDICP	127	AI/O
P2MDICN	128	AI/O

7. Pin Descriptions

7.1. Media Dependent Interface Pins

Table 2. Media Dependent Interface Pins

Pin Name	Pin No.	Type	Drive (mA)	Description
P0MDIAP/N	99	AI/O	10	Port 0 Media Dependent Interface A~D. For 1000Base-T operation, differential data from the media is transmitted and received on all four pairs. For 100Base-Tx and 10Base-T operation, only MDIAP/N and MDIBP/N are used. Auto MDIX can reverse the pairs MDIAP/N and MDIBP/N. Each of the differential pairs has an internal 100-ohm termination resistor.
	100			
P0MDIBP/N	101			
	102			
P0MDICP/N	104			
	105			
P0MDIDP/N	106	AI/O	10	Port 1 Media Dependent Interface A~D. For 1000Base-T operation, differential data from the media is transmitted and received on all four pairs. For 100Base-Tx and 10Base-T operation, only MDIAP/N and MDIBP/N are used. Auto MDIX can reverse the pairs MDIAP/N and MDIBP/N. Each of the differential pairs has an internal 100-ohm termination resistor.
	107			
P1MDIAP/N	110			
	111			
P1MDIBP/N	112			
	113			
P1MDICP/N	115	AI/O	10	Port 2 Media Dependent Interface A~D. For 1000Base-T operation, differential data from the media is transmitted and received on all four pairs. For 100Base-Tx and 10Base-T operation, only MDIAP/N and MDIBP/N are used. Auto MDIX can reverse the pairs MDIAP/N and MDIBP/N. Each of the differential pairs has an internal 100-ohm termination resistor.
	116			
P1MDIDP/N	117			
	118			
P2MDIAP/N	122			
	123			
P2MDIBP/N	124	AI/O	10	Port 2 Media Dependent Interface A~D. For 1000Base-T operation, differential data from the media is transmitted and received on all four pairs. For 100Base-Tx and 10Base-T operation, only MDIAP/N and MDIBP/N are used. Auto MDIX can reverse the pairs MDIAP/N and MDIBP/N. Each of the differential pairs has an internal 100-ohm termination resistor.
	125			
P2MDICP/N	127			
	128			
P2MDIDP/N	1			
	2			

Pin Name	Pin No.	Type	Drive (mA)	Description
P3MDIAP/N	4	AI/O	10	Port 3 Media Dependent Interface A~D. For 1000Base-T operation, differential data from the media is transmitted and received on all four pairs. For 100Base-Tx and 10Base-T operation, only MDIAP/N and MDIBP/N are used. Auto MDIX can reverse the pairs MDIAP/N and MDIBP/N. Each of the differential pairs has an internal 100-ohm termination resistor.
	5			
P3MDIBP/N	6			
	7			
P3MDICP/N	10			
	11			
P3MDIDP/N	12			
	13			
P4MDIAP/N	31	AI/O	10	Port 4 Media Dependent Interface A~D. For 1000Base-T operation, differential data from the media is transmitted and received on all four pairs. For 100Base-Tx and 10Base-T operation, only MDIAP/N and MDIBP/N are used. Auto MDIX can reverse the pairs MDIAP/N and MDIBP/N. Each of the differential pairs has an internal 100-ohm termination resistor.
	32			
P4MDIBP/N	33			
	34			
P4MDICP/N	37			
	38			
P4MDIDP/N	39			
	40			

7.2. LED Pins

LED0, LED1, and LED2 of Port n indicate information that can be defined via register or EEPROM. When the LED pin is pulled low, the LED output polarity will be high active. When the LED pin is pulled high, the LED output polarity will change from high active to low active. See section 9.19 LED Indicator, page 32 for more details.

Table 3. LED Pins

Pin Name	Pin No.	Type	Drive (mA)	Description
P4LED2	72	I/O _{PU}	-	Port 4 LED2 Output Signal. P4LED2 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P4LED1	74	I/O _{PU}	-	Port 4 LED1 Output Signal. P4LED1 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P4LED0/ EEPROM_MOD	73	I/O _{PU}	-	Port 4 LED0 Output Signal. P4LED0 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P3LED2	76	I/O _{PU}	-	Port 3 LED2 Output Signal. P3LED2 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P3LED1	75	I/O _{PU}	-	Port 3 LED1 Output Signal. P3LED1 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P3LED0/ RESERVED	77	I/O _{PU}	-	Port 3 LED0 Output Signal. P3LED0 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.

Pin Name	Pin No.	Type	Drive (mA)	Description
P2LED2/DIS_8051	78	I/O _{PU}	-	Port 2 LED2 Output Signal. P2LED2 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details
P2LED1	80	I/O _{PU}	-	Port 2 LED1 Output Signal. P2LED1 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P2LED0/ DISAUTOLOAD	79	I/O _{PU}	-	Port 2 LED0 Output Signal. P2LED0 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P1LED2	81	I/O _{PU}	-	Port 1 LED2 Output Signal. P1LED2 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P1LED1	83	I/O _{PU}	-	Port 1 LED1 Output Signal. P1LED1 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P1LED0	82	I/O _{PU}	-	Port 1 LED0 Output Signal. P1LED0 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P0LED2/ EN_PHY	84	I/O _{PU}	-	Port 0 LED2 Output Signal. P0LED2 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P0LED1/LED_DA	88	I/O _{PU}	-	Port 0 LED1 Output Signal. P0LED1 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.
P0LED0/LED_CK/ SMI_SEL	89	I/O _{PU}	-	Port 0 LED0 Output Signal. P0LED0 indicates information is defined by register or EEPROM. See section 9.19 LED Indicator, page 32 for more details.

7.3. Configuration Strapping Pins

Table 4. Configuration Strapping Pins

Pin Name	Pin No.	Type	Description
EEPROM_MOD/ P4LED0	73	I/O _{PU}	EEPROM Mode Selection. Pull Up: EEPROM 24Cxx Size great than 16Kbits (24C32~24C256) Pull Down: EEPROM 24Cxx Size less than or equal to 16Kbit (24C02~24C16). <i>Note: This pin must be kept floating, or pulled high or low via an external 4.7k ohm resistor upon power on or reset.</i> <i>When this pin is pulled low, the LED output polarity will be high active. When this pin is pulled high, the LED output polarity will change from high active to low active. See section 9.19 LED Indicator, page 32 for more details.</i>

Pin Name	Pin No.	Type	Description
RESERVED/P3LED0	77	I/O _{PU}	Internal Use/Reserved. <i>Note: This pin must be pulled low via an external 4.7k ohm resistor upon power on or reset for normal operation.</i> <i>When pulled low, the LED output polarity will be high active. See section 9.19 LED Indicator, page 32 for more details.</i>
DIS_8051/P2LED2	78	I/O _{PU}	Disable Embedded 8051. Pull Up: Disable embedded 8051 upon power on or reset Pull Down: Enable embedded 8051 upon power on or reset <i>Note: This pin must be kept floating, or pulled high or low via an external 4.7k ohm resistor upon power on or reset.</i> <i>When this pin is pulled low, the LED output polarity will be high active. When this pin is pulled high, the LED output polarity will change from high active to low active. See section 9.19 LED Indicator, page 32 for more details.</i>
DISAUTOLOAD/ P2LED0	79	I/O _{PU}	Disable EEPROM Autoload. Pull Up: Disable EEPROM autoload upon power on or reset Pull Down: Enable EEPROM autoload upon power on or reset <i>Note1: When DIS_8051 = 1 and DISAUTOLOAD = 0, the EEPROM data will be treat as register configuration data upon power on or reset initial stage. When DIS_8051 = 0 and DISAUTOLOAD = 0, the EEPROM data will be loaded to embedded 8051 instruction memory upon power on or reset.</i> <i>Note2: This pin must be kept floating, or pulled high or low via an external 4.7k ohm resistor upon power on or reset.</i> <i>When this pin is pulled low, the LED output polarity will be high active. When this pin is pulled high, the LED output polarity will change from high active to low active. See section 9.19 LED Indicator, page 32 for more details.</i>
EN_PHY/P0LED2	84	I/O _{PU}	Enable Embedded PHY. Pull Up: Enable embedded PHY Pull Down: Disable embedded PHY <i>Note: This pin must be kept floating, or pulled high or low via an external 4.7k ohm resistor upon power on or reset.</i> <i>When pulled high, the LED output polarity will be low active. See section 9.19 LED Indicator, page 32 for more details.</i>
SMI_SEL/P0LED0/LED_ CK	89	I/O _{PU}	EEPROM SMI/MII Management Interface Selection. Pull Up: EEPROM SMI interface Pull Down: MII Management Interface <i>Note: This pin must be kept floating, or pulled high or low via an external 4.7k ohm resistor upon power on or reset.</i> <i>When pulled high, the LED output polarity will be low active. See section 9.19 LED Indicator, page 32 for more details.</i>
BUZZER/DIS_LPD	25	I/O _{PU}	Realtek Loop Detection Configuration. Pull Up: Disable Loop detection function Pull Down: Enable Loop detection function <i>Note: This pin must be kept floating, or pulled high or low via an external 4.7k ohm resistor upon power on or reset. 2KHz signal out when looping is detected.</i>

7.4. Miscellaneous Pins

Table 5. Miscellaneous Pins

Pin Name	Pin No.	Type	Description
XTALI	94	AI	25MHz Crystal Clock Input and Feedback Pin. 25MHz +/-50ppm tolerance crystal reference or oscillator input.
XTALO	93	AO	25MHz Crystal Clock Output Pin. 25MHz +/-50ppm tolerance crystal output.
MDIREF	16	AO	Reference Resistor. A 2.49K ohm (1%) resistor must be connected between MDIREF and GND.
RESERVED	51	AO	Reserved. Must be left floating in normal operation.
RESERVED	52	AO	Reserved. Must be left floating in normal operation.
SCK/MMD_MDC	96	I/O _{PU}	EEPROM SMI Interface Clock/MII Management Interface Clock (selected via the hardware strapping pin 89, SMI_SEL).
SDA/MMD_MDIO	97	I/O _{PU}	EEPROM SMI Interface Data/MII Management Interface Data (selected via the hardware strapping pin 89, SMI_SEL).
nRESET	95	I _S	System Reset Input Pin. When low active will reset the RTL8367.
RESERVED	54	I/O	Reserved. Must be pulled low with 1k ohm resistor when in normal operation.
RESERVED	55	I/O	Reserved. Must be pulled low with 1k ohm resistor when in normal operation.
RESERVED	57	O	Reserved. Must be left floating in normal operation.
RESERVED	58	I/O	Reserved. Must be pulled low with 1k ohm resistor when in normal operation.
RESERVED	60	I/O	Reserved. Must be pulled low with 1k ohm resistor when in normal operation.
RESERVED	61	O	Reserved. Must be pulled low with 1k ohm resistor when in normal operation.
RESERVED	62	I/O	Reserved. Must be pulled low with 1k ohm resistor when in normal operation.

7.5. Test Pins

Table 6. Test Pins

Pin Name	Pin No.	Type	Description
RTT1	18	AO	Reserved for Internal Use. Must be left floating.
RTT2	19	AO	Reserved for Internal Use. Must be left floating.
ATESTCK1	44	AO	Reserved for Internal Use. Must be left floating.
TEST_MODE	56	I _{PD}	Enable TEST Mode. <i>Note: This pin must be kept floating, or pulled low via an external 4.7k ohm resistor upon power on or reset for normal application.</i>

7.6. Power and GND Pins

Table 7. Power and GND Pins

Pin Name	Pin No.	Type	Description
DVDDIO	21, 53, 68	P	Digital I/O High Voltage Power for LED, SMI, nRESET.
DVDDL	26, 66, 71, 91	P	Digital Low Voltage Power.
AVDDH	3, 14, 20, 30, 41, 50, 92, 98, 108	AP	Analog High Voltage Power.
AVDDL	8, 17, 36, 46, 103, 114, 126	AP	Analog Low Voltage Power.
PLLVDDL0	119	AP	PLL0 Low Voltage Power.
PLLVDDL1	43	AP	PLL1 Low Voltage Power.
GND	9, 23, 24, 29, 35, 42, 47, 48, 49, 63, 69, 70, 86, 90, 109, 121	G	GND.
AGND	15	AG	Analog GND.
PLLGND0	120	AG	PLL0 GND.
PLLGND1	45	AG	PLL1 GND.

7.7. NC (No Connection) Pins

Table 8. NC (No Connection) Pin

Pin Name	Pin No.	Type	Description
NC	22, 27, 28, 59, 64, 65, 67, 85, 87	-	No Connection.

8. Physical Layer Functional Overview

8.1. MDI Interface

The RTL8367 embeds five Gigabit Ethernet PHYs in one chip. Each port uses a single common MDI interface to support 1000Base-T, 100Base-Tx, and 10Base-T. This interface consists of four signal pairs- A, B, C, and D. Each signal pair consists of two bi-directional pins that can transmit and receive at the same time. The MDI interface has internal termination resistors, and therefore reduces BOM cost and PCB complexity. For 1000Base-T, all four pairs are used in both directions at the same time. For 10/100 links and during auto-negotiation, only pairs A and B are used.

8.2. 1000Base-T Transmit Function

The 1000Base-TX transmit function performs 8B/10B coding, scrambling, and 4D-PAM5 encoding. These code groups are passed through a waveform-shaping filter to minimize EMI effects, and are transmitted onto 4-pair CAT5 cable at 125MBaud/s through a D/A converter.

8.3. 1000Base-T Receive Function

Input signals from the media pass through the sophisticated on-chip hybrid circuit to subtract the transmitted signal from the input signal for effective reduction of near-end echo. The received signal is then processed with state-of-the-art technology, e.g., adaptive equalization, BLW (Baseline Wander) correction, cross-talk cancellation, echo cancellation, timing recovery, error correction, and 4D-PAM5 decoding. The 8-bit-wide data is recovered and is sent to the GMII interface at a clock speed of 125MHz. The RX MAC retrieves the packet data from the internal receive MII/GMII interface and sends it to the packet buffer manager.

8.4. 100Base-TX Transmit Function

The 100Base-TX transmit function performs parallel to serial conversion, 4B/5B coding, scrambling, NRZ/NRZI conversion, and MLT-3 encoding. The 5-bit serial data stream after 4B/5B coding is then scrambled as defined by the TP-PMD Stream Cipher function to flatten the power spectrum energy such that EMI effects can be reduced significantly.

The scrambled seed is based on PHY addresses and is unique for each port. After scrambling, the bit stream is driven onto the network media in the form of MLT-3 signaling. The MLT-3 multi-level signaling technology moves the power spectrum energy from high frequency to low frequency, which also reduces EMI emissions.

8.5. 100Base-TX Receive Function

The receive path includes a receiver composed of an adaptive equalizer and DC restoration circuits (to compensate for an incoming distorted MLT-3 signal), an MLT-3 to NRZI and NRZI to NRZ converter to convert analog signals to digital bit-stream, and a PLL circuit to clock data bits with minimum bit error rate. A de-scrambler, 5B/4B decoder, and serial-to-parallel conversion circuits are followed by the PLL circuit. Finally, the converted parallel data is fed into the MAC.

8.6. 10Base-T Transmit Function

The output 10Base-T waveform is Manchester-encoded before it is driven onto the network media. The internal filter shapes the driven signals to reduce EMI emissions, eliminating the need for an external filter.

8.7. 10Base-T Receive Function

The Manchester decoder converts the incoming serial stream to NRZ data when the squelch circuit detects the signal level is above squelch level.

8.8. Auto-Negotiation for UTP

The RTL8367 obtains the states of duplex, speed, and flow control ability for each port in UTP mode through the auto-negotiation mechanism defined in the IEEE 802.3 specifications. During auto-negotiation, each port advertises its ability to its link partner and compares its ability with advertisements received from its link partner. By default, the RTL8367 advertises full capabilities (1000Full, 100Full, 100Half, 10Full, 10Half) together with flow control ability.

8.9. Crossover Detection and Auto Correction

The RTL8367 automatically determines whether or not it needs to crossover between pairs (see Table 9) so that an external crossover cable is not required. When connecting to another device that does not perform MDI crossover, when necessary, the RTL8367 automatically switches its pin pairs to communicate with the remote device. When connecting to another device that does have MDI crossover capability, an algorithm determines which end performs the crossover function.

The crossover detection and auto correction function can be disabled via register configuration. The pin mapping in MDI and MDI Crossover mode is given below.

Table 9. Media Dependent Interface Pin Mapping

Pairs	MDI			MDI Crossover		
	1000Base-T	100Base-TX	10Base-T	1000Base-T	100Base-TX	10Base-T
A	A	TX	TX	B	RX	RX
B	B	RX	RX	A	TX	TX
C	C	Unused	Unused	D	Unused	Unused
D	D	Unused	Unused	C	Unused	Unused

8.10. Polarity Correction

The RTL8367 automatically corrects polarity errors on the receiver pairs in 1000Base-T and 10Base-T modes. In 100Base-TX mode, the polarity is irrelevant.

In 1000Base-T mode, receive polarity errors are automatically corrected based on the sequence of idle symbols. Once the de-scrambler is locked, the polarity is also locked on all pairs. The polarity becomes unlocked only when the receiver loses lock.

In 10Base-T mode, polarity errors are corrected based on the detection of valid spaced link pulses. The detection begins during the MDI crossover detection phase and locks when the 10Base-T link is up. The polarity becomes unlocked when the link is down.

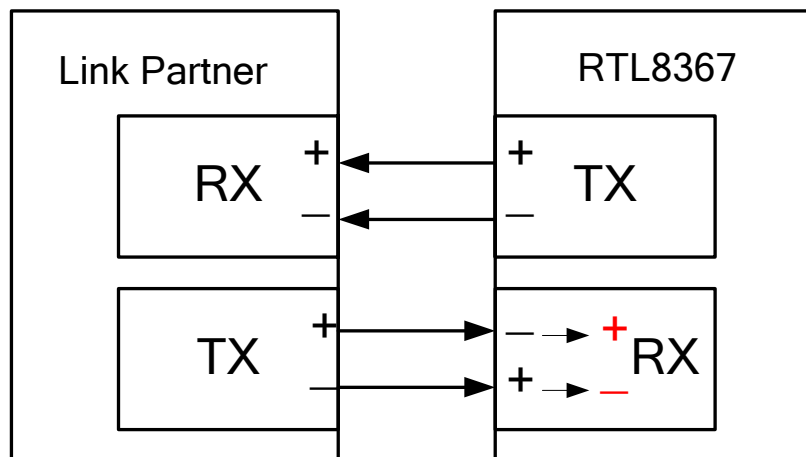


Figure 4. Conceptual Example of Polarity Correction

9. General Function Description

9.1. *Reset*

9.1.1. Hardware Reset

In a power-on reset, an internal power-on reset pulse is generated and the RTL8367 will start the reset initialization procedures. These are:

- Determine various default settings via the hardware strap pins at the end of the nRESET signal
- Autoload the configuration from EEPROM if EEPROM is detected
- Complete the embedded SRAM BIST process
- Initialize the packet buffer descriptor allocation
- Initialize the internal registers and prepare them to be accessed by the external CPU

9.1.2. Software Reset

The RTL8367 supports two software resets; a chip reset and a soft reset.

9.1.2.1 *CHIP_RESET*

When CHIP_RESET is set to 0b1 (write and self-clear), the chip will take the following steps:

1. Download configuration from strap pin and EEPROM
2. Start embedded SRAM BIST (Built-In Self Test)
3. Clear all the Lookup and VLAN tables
4. Reset all registers to default values
5. Restart the auto-negotiation process

9.1.2.2 *SOFT_RESET*

When SOFT_RESET is set to 0b1 (write and self-clear), the chip will take the following steps:

1. Clear the FIFO and re-start the packet buffer link list
2. Restart the auto-negotiation process

9.2. *IEEE 802.3x Full Duplex Flow Control*

The RTL8367 supports IEEE 802.3x flow control in 10/100/1000M modes. Flow control can be decided in two ways:

- When Auto-Negotiation is enabled, flow control depends on the result of NWay
- When Auto-Negotiation is disabled, flow control depends on register definition

9.3. Half Duplex Flow Control

In half duplex mode, the CSMA/CD media access method is the means by which two or more stations share a common transmission medium. To transmit, a station waits (defers) for a quiet period on the medium (that is, no other station is transmitting) and then sends the intended message in bit-serial form. If the message collides with that of another station, then each transmitting station intentionally transmits for an additional predefined period to ensure propagation of the collision throughout the system. The station remains silent for a random amount of time (backoff) before attempting to transmit again.

When a transmission attempt has terminated due to a collision, it is retried until it is successful. The scheduling of the retransmissions is determined by a controlled randomization process called “truncated binary exponential backoff”. At the end of enforcing a collision (jamming), the switch delays before attempting to retransmit the frame. The delay is an integer multiple of slot time (512 bit times). The number of slot times to delay before the *n*th retransmission attempt is chosen as a uniformly distributed random integer ‘*r*’ in the range:

$$0 \leq r < 2k$$

where:

$k = \min(n, \text{backoffLimit})$. The backoffLimit for the RTL8367 is 9.

The half duplex back-off algorithm in the RTL8367 does not have the maximum retry count limitation of 16 (as defined in IEEE 802.3). This means packets in the switch will not be dropped if the back-off retry count is over 16.

9.3.1. Back-Pressure Mode

In Back-Pressure mode, the RTL8367 sends a 4-byte jam pattern (data=0xAA) to collide with incoming packets when congestion control is activated. The Jam pattern collides at the fourth byte counted from the preamble. The RTL8367 supports 48PASS1, which receives one packet after 48 consecutive jam collisions (data collisions are not included in the 48). Enable this function to prevent port partition after 63 consecutive collisions (data collisions + consecutive jam collisions).

9.4. Search and Learning

Search

When a packet is received, the RTL8367 uses the destination MAC address, Filtering Identifier (FID) and enhanced Filtering Identifier (FID) to search the 2K-entry look-up table. The 48-bit MAC address, 4-bit FID and 3-bit EFID use a hash algorithm, to calculate an 11-bit index value. The RTL8367 uses the index to compare the packet MAC address with the entries (MAC addresses) in the look-up table. This is the 'Address Search'. If the destination MAC address is not found, the switch will broadcast the packet according to VLAN configuration.

Learning

The RTL8367 uses the source MAC address, FID, and EFID of the incoming packet to hash into a 9-bit index. It then compares the source MAC address with the data (MAC addresses) in this index. If there is a match with one of the entries, the RTL8367 will update the entry with new information. If there is no match and the 2K entries are not all occupied by other MAC addresses, the RTL8367 will record the source MAC address and ingress port number into an empty entry. This process is called 'Learning'.

The RTL8367 supports a 64-entry Content Addressable Memory (CAM) to avoid look-up table hash collisions. When all 2K entries in the look-up table index are occupied, the source MAC address can be learned into the 64-entry CAM. If both the look-up table and the CAM are full, the source MAC address will not be learned in the RTL8367.

Address aging is used to keep the contents of the address table correct in a dynamic network topology. The look-up engine will update the time stamp information of an entry whenever the corresponding source MAC address appears. An entry will be invalid (aged out) if its time stamp information is not refreshed by the address learning process during the aging time period. The aging time of the RTL8367 is between 200 and 400 seconds (typical is 300 seconds).

9.5. SVL and IVL/SVL

The RTL8367 supports a 16-group Filtering Identifier (FID) for L2 search and learning. In default operation, all VLAN entries belong to the same FID. This is called Shared VLAN Learning (SVL). If VLAN entries are configured to different FIDs, then the same source MAC address with multiple FIDs can be learned into different look-up table entries. This is called Independent VLAN Learning and Shared VLAN Learning (IVL/SVL).

9.6. Illegal Frame Filtering

Illegal frames such as CRC error packets, runt packets (length <64 bytes), and oversize packets (length >maximum length) will be discarded by the RTL8367. The maximum packet length may be set to 1522, 1536, 1552, or 16K bytes.

9.7. IEEE 802.3 Reserved Group Addresses Filtering Control

The RTL8367 supports the ability to drop/forward IEEE 802.3 specified reserved group MAC addresses: 01-80-C2-00-00-00 to 01-80-C2-00-00-2F. The default setting enables forwarding of these reserved group MAC address control frames. Frames with group MAC address 01-80-C2-00-00-01 (802.3x Pause) and 01-80-C2-00-00-02 (802.3ad LACP) will always be filtered. Table 10 shows the Reserved Multicast Address (RMA) configuration mode from 01-80-C2-00-00-00 to 01-80-C2-00-00-2F.

Table 10. Reserved Multicast Address Configuration Table

Assignment	Value
Bridge Group Address	01-80-C2-00-00-00
IEEE Std 802.3, 1988 Edition, Full Duplex PAUSE operation	01-80-C2-00-00-01
IEEE Std 802.3ad Slow Protocols-Multicast address	01-80-C2-00-00-02
IEEE Std 802.1X PAE address	01-80-C2-00-00-03
Provider Bridge group address	1-80-C2-00-00-08
Undefined 802.1 Address	01-80-C2-00-00-04 ~ 01-80-C2-00-00-07 & 01-80-C2-00-00-09 ~ 01-80-C2-00-00-0C & 01-80-C2-00-00-0F
Provider Bridge MVRP address	01-80-C2-00-00-0D
IEEE Std 802.1AB Link Layer Discovery Protocol address	01-80-C2-00-00-0E
All LANs Bridge Management Group Address	01-80-C2-00-00-10
Load Server Generic Address	01-80-C2-00-00-11
Loadable Device Generic Address	01-80-C2-00-00-12
Undefined 802.1 Address	01-80-C2-00-00-13 ~ 01-80-C2-00-00-17 & 01-80-C2-00-00-19 & 01-80-C2-00-00-1B ~ 01-80-C2-00-00-1F
Generic Address for All Manager Stations	01-80-C2-00-00-18
Generic Address for All Agent Stations	01-80-C2-00-00-1a
GMRP Address	01-80-C2-00-00-20
GVRP address	01-80-C2-00-00-21
Undefined GARP Address	01-80-C2-00-00-22 01-80-C2-00-00-2F

9.8. Broadcast/Multicast/Unknown DA Storm Control

The RTL8367 enables or disables per-port broadcast/multicast/unknown DA storm control by setting registers (default is disabled). After the receiving rate of broadcast/multicast/unknown DA packets

exceeds a reference rate, all other broadcast/multicast/unknown DA packets will be dropped. The reference rate is set via register configuration.

9.9. Port Security Function

The RTL8367 supports three types of security function to prevent malicious attacks.

- Per-port enable/disable SA auto-learning for an ingress packet
- Per-port enable/disable look-up table aging update function for an ingress packet
- Per-port enable/disable drop all unknown DA packets

9.10. MIB Counters

The RTL8367 supports a set of counters to support management functions.

- MIB-II (RFC 1213)
- Ethernet-Like MIB (RFC 3635)
- Interface Group MIB (RFC 2863)
- RMON (RFC 2819)
- Bridge MIB (RFC 1493)
- Bridge MIB Extension (RFC 2674)

9.11. Port Mirroring

The RTL8367 supports one set of port mirroring functions for all ports. The TX, or RX, or both TX/RX packets of the source port can be monitored to multiple mirror ports.

9.12. VLAN Function

The RTL8367 supports 4K VLAN groups. These can be configured as port-based VLANs, IEEE 802.1Q tag-based VLANs, and Protocol-based VLANs. Two ingress-filtering and egress-filtering options provide flexible VLAN configuration:

Ingress Filtering

- The acceptable frame type of the ingress process can be set to ‘Admit All’ or ‘Admit All Tagged’
- ‘Admit’ or ‘Discard’ frames associated with a VLAN for which that port is not in the member set

Egress Filtering

- ‘Forward’ or ‘Discard’ Leaky VLAN frames between different VLAN domains
- ‘Forward’ or ‘Discard’ Multicast VLAN frames between different VLAN domains

The VLAN tag can be inserted or removed at the output port. The RTL8367 will insert a Port VID (PVID) for untagged frames, or remove the tag from tagged frames. The RTL8367 also supports a special insert VLAN tag function to separate traffic from the WAN and LAN sides in Router and Gateway applications.

In router applications, the router may want to know which input port this packet came from. The RTL8367 supports Port VID (PVID) for each port and can insert a PVID in the VLAN tag on egress. Using this function, VID information carried in the VLAN tag will be changed to PVID. The RTL8367 also provides an option to admit VLAN tagged packets with a specific PVID only. If this function is enabled, it will drop non-tagged packets and packets with an incorrect PVID.

9.12.1. Port-Based VLAN

This default configuration of the VLAN function can be modified via an attached serial EEPROM or EEPROM SMI Slave interface. The 4K-entry VLAN Table designed into the RTL8367 provides full

flexibility for users to configure the input ports to associate with different VLAN groups. Each input port can join with more than one VLAN group.

Port-based VLAN mapping is the simplest implicit mapping rule. Each ingress packet is assigned to a VLAN group based on the input port. It is not necessary to parse and inspect frames in real-time to determine their VLAN association. All the packets received on a given input port will be forwarded to this port's VLAN members.

9.12.2. IEEE 802.1Q Tag-Based VLAN

The RTL8367 supports 4K VLAN entries to perform 802.1Q tag-based VLAN mapping. In 802.1Q VLAN mapping, the RTL8367 uses a 12-bit explicit identifier in the VLAN tag to associate received packets with a VLAN. The RTL8367 compares the explicit identifier in the VLAN tag with the 4K VLAN Table to determine the VLAN association of this packet, and then forwards this packet to the member set of that VLAN. Two VIDs are reserved for special purposes. One of them is all 1's, which is reserved and currently unused. The other is all 0's, which indicates a priority tag. A priority-tagged frame should be treated as an untagged frame.

When '802.1Q tag aware VLAN' is enabled, the RTL8367 performs 802.1Q tag-based VLAN mapping for tagged frames, but still performs port-based VLAN mapping for untagged frames. If '802.1Q tag aware VLAN' is disabled, the RTL8367 performs only port-based VLAN mapping both on non-tagged and tagged frames. The processing flow when '802.1Q tag aware VLAN' is enabled is illustrated below.

Two VLAN ingress filtering functions are supported in registers by the RTL8367. One is the 'VLAN tag admit control', which provides the ability to receive VLAN-tagged frames only. Untagged or priority tagged (VID=0) frames will be dropped. The other is 'VLAN member set ingress filtering', which will drop frames if the ingress port is not in the member set.

9.12.3. Protocol-Based VLAN

The RTL8367 supports a 4-group Protocol-based VLAN configuration. The packet format can be RFC 1042, LLC, or Ethernet, as shown in Figure 5. There are 4 configuration tables to assign the frame type and corresponding field value. Taking IP packet configuration as an example, the user can configure the frame type to be 'Ethernet', and value to be '0x0800'. Each table will index to one of the entries in the 4K-entry VLAN table. The packet stream will match the protocol type and the value will follow the VLAN member configuration of the indexed entry to forward the packets.

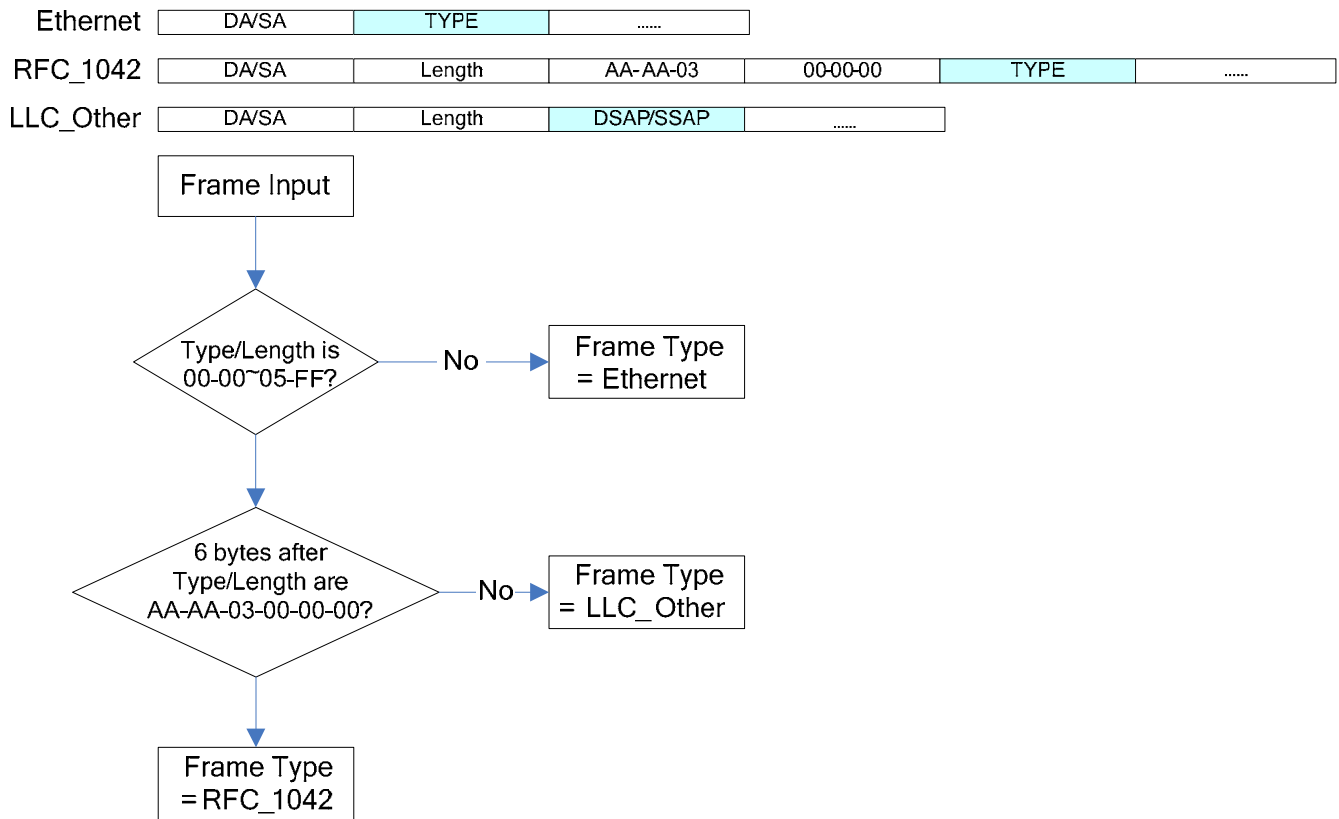


Figure 5. Protocol-Based VLAN Frame Format and Flow Chart

9.12.4. Port VID

In a router application, the router may want to know which input port this packet came from. The RTL8367 supports Port VID (PVID) for each port to insert a PVID in the VLAN tag for untagged or priority tagged packets on egress. When 802.1Q tag-aware VLAN is enabled, VLAN tag admit control is enabled, and non-PVID Discard is enabled at the same time. When these functions are enabled, the RTL8367 will drop non-tagged packets and packets with an incorrect PVID.

9.13. QoS Function

The RTL8367 supports 8 priority queues and input bandwidth control. Packet priority selection can depend on Port-based priority, 802.1p/Q Tag-based priority, IPv4/IPv6 DSCP-based priority, and ACL-based priority. When multiple priorities are enabled in the RTL8367, the packet's priority will be assigned based on the priority selection table.

Each queue has one leaky bucket for Average Packet Rate. Per-queue in each output port can be set as Strict Priority (SP) or Weighted Fair Queue (WFQ) for packet scheduling algorithm.

9.13.1. Input Bandwidth Control

Input bandwidth control limits the input bandwidth. When input traffic is more than the RX Bandwidth parameter, this port will either send out a 'pause ON' frame, or drop the input packet depending on register setup. Per-port input bandwidth control rates can be set from 8Kbps to 1Gbps (in 8Kbps steps).

9.13.2. Priority Assignment

Priority assignment specifies the priority of a received packet according to various rules. The RTL8367 can recognize the QoS priority information of incoming packets to give a different egress service priority.

The RTL8367 identifies the priority of packets based on several types of QoS priority information:

- Port-based priority
- 802.1p/Q-based priority
- IPv4/IPv6 DSCP-based priority
- ACL-based priority
- VLAN based priority
- MAC based priority
- SVLAN based priority

9.13.3. Priority Queue Scheduling

The RTL8367 supports MAX-MIN packet scheduling.

Packet scheduling offers two modes:

- APR leaky bucket, which specifies the average rate of one queue
- Weighted Fair Queue (WFQ), which decides which queue is selected in one slot time to guarantee the minimal packet rate of one queue

In addition, each queue of each port can select Strict Priority or WFQ packet scheduling according to packet scheduling mode. Figure 6 shows the RTL8367 packet-scheduling diagram.

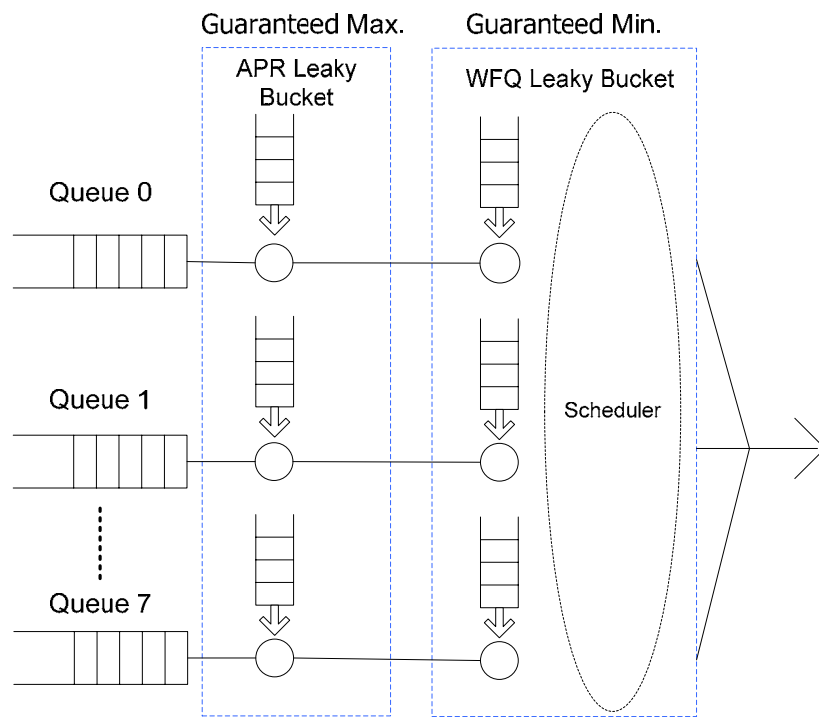


Figure 6. MAX-MIN Scheduling Diagram

9.13.4. IEEE 802.1p/Q and DSCP Remarking

The RTL8367 supports the IEEE 802.1p/Q and IP DSCP (Differentiated Services Code Point) remarking function. When packets egress from one of the 8 queues, the packet's 802.1p/Q priority and IP DSCP can optionally be remarked to a configured value. Each output queue has a 3-bit 802.1p/Q, and a 6-bit IP DSCP value configuration register.

9.13.5. ACL-Based Priority

The RTL8367 supports 64-entry ACL (Access Control List) rules. When a packet is received, its physical port, Layer2, Layer3, and Layer4 information are recorded and compared to ACL entries.

If a received packet matches multiple entries, the entry with the lowest address is valid. If the entry is valid, the action bit and priority bit will be applied.

- If the action bit is ‘Drop’, the packet will be dropped. If the action bit is ‘CPU’, the packet will be trapped to the CPU instead of forwarded to non-CPU ports (except where it will be dropped by rules other than the ACL rule)
- If the action bit is ‘Permit’, ACL rules will override other rules
- If the action bit is ‘Mirror’, the packet will be forwarded to the mirror port and the L2 lookup result destination port. The mirror port indicates the port configured in the port mirror mechanism
- The priority bit will take effect only if the action bit is ‘CPU’, ‘Permit’, and ‘Mirror’. The Priority bit is used to determine the packet queue ID according to the priority assignment mechanism

9.14. IGMP & MLD Snooping Function

The RTL8367 supports H/W IGMP v1/v2/v3 and MLD v1/v2 snooping. The RTL8367 can learn multicast group membership information automatically without software effort. Multicast data packet would be forwarded to member ports only. The IGMP & MLD Snooping function in the RTL8367 also supports “Fast Leave” for those application which wants to remove membership after receiving a Leave packet.

9.15. IEEE 802.1x Function

The RTL8367 supports IEEE 802.1x Port-based/MAC-based Access Control.

- Port-Based Access Control for each port
- Authorized Port-Based Access Control for each port
- Port-Based Access Control Direction for each port
- MAC-Based Access Control for each port
- MAC-Based Access Control Direction
- Optional Unauthorized Behavior
- Guest VLAN

9.15.1. Port-Based Access Control

Each port of the RTL8367 can be set to 802.1x port-based authenticated checking function usage and authorized status. Ports with 802.1X unauthorized status will drop received/transmitted frames.

9.15.2. Authorized Port-Based Access Control

If a dedicated port is set to 802.1x port-based access control, and passes the 802.1x authorization, then its port authorization status can be set to authorized.

9.15.3. Port-Based Access Control Direction

Ports with 802.1X unauthorized status will drop received/transmitted frames only when port authorization direction is 'BOTH'. If the authorization direction of an 802.1X unauthorized port is IN, incoming frames to that port will be dropped, but outgoing frames will be transmitted.

9.15.4. MAC-Based Access Control

MAC-Based Access Control provides authentication for multiple logical ports. Each logical port represents a source MAC address. There are multiple logical ports for a physical port. When a logical port or a MAC address is authenticated, the relevant source MAC address has the authorization to access the network. A frame with a source MAC address that is not authenticated by the 802.1x function will be dropped or trapped to the CPU.

9.15.5. MAC-Based Access Control Direction

Unidirectional and Bi-directional control are two methods used to process frames in 802.1x. As the system cannot predict which port the DA is on, a system-wide MAC-based access control direction setup is provided for determining whether receiving or bi-direction must be authorized.

If MAC-based access control direction is BOTH, then received frames with unauthenticated SA or unauthenticated DA will be dropped. When MAC-based access control direction is IN, only received frames with unauthenticated SA will be dropped.

9.15.6. Optional Unauthorized Behavior

Both in Port-Based Network Access Control and MAC-Based Access Control, a whole system control setup is provided to determine unauthorized frame dropping, trapping to CPU, or tagging as belonging to a Guest VLAN (see the following 'Guest VLAN' section).

9.15.7. Guest VLAN

When the RTL8367 enables the Port-based or MAC-based 802.1x function, and the connected PC does not support the 802.1x function or does not pass the authentication procedure, the RTL8367 will drop all packets from this port.

The RTL8367 also supports one Guest VLAN to allow unauthorized ports or packets to be forwarded to a limited VLAN domain. The user can configure one VLAN ID and member set for these unauthorized packets.

9.16. IEEE 802.1D Function

When using IEEE 802.1D, the RTL8367 supports 16 sets and four status' for each port for CPU implementation 802.1D (STP) and 802.1s (MSTP) function:

- Disabled: The port will not transmit/receive packets, and will not perform learning
- Blocking: The port will only receive BPDU spanning tree protocol packets, but will not transmit any packets, and will not perform learning
- Learning: The port will receive any packet, including BPDU spanning tree protocol packets, and will perform learning, but will only transmit BPDU spanning tree protocol packets
- Forwarding: The port will transmit/receive all packets, and will perform learning

The RTL8367 also supports a per-port transmission/reception enable/disable function. Users can control the port state via register.

9.17. Embedded 8051

An 8051 MCU is embedded in the RTL8367 to support management functions. The 8051 MCU can access all of the registers in the RTL8367 through the internal bus. With the Network Interface Circuit (NIC) acting as the data path, the 8051 MCU connects to the switch core and can transmit frames to or receive frames from the Ether network. The features of the 8051 MCU are listed below:

- 256 Bytes fast internal RAM
- On-chip 32K data memory
- On-chip 16K code memory
- Supports code-banking
- 12KBytes NIC buffer
- EEPROM read/write ability

9.18. Realtek Cable Test (RTCT)

The RTL8367 physical layer transceivers use DSP technology to implement the Realtek Cable Test (RTCT) feature. The RTCT function can be used to detect short, open, or impedance mismatch in each differential pair. The RTL8367 also provides LED support to indicate test status and results.

9.19. LED Indicator

The RTL8367 supports parallel LEDs for each port. Each port has three LED indicator pins, LED0, LED1, and LED2. Each pin may have different indicator information (defined in Table 11). Refer to section 7.2 LED Pins, page 11 for pin details. Upon reset, the RTL8367 supports chip diagnostics and LED operation test by blinking all LEDs once.

Table 11. LED Definitions

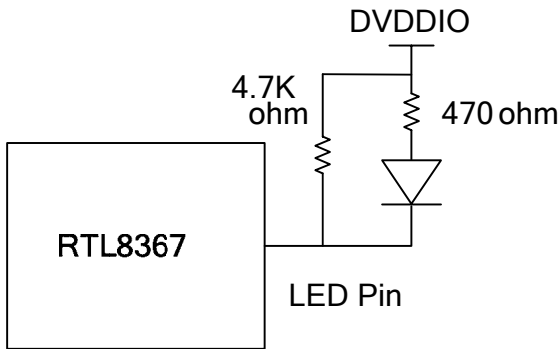
LED Statuses	Description
LED_Off	LED pin Output disable.
Dup/Col	Duplex/Collision, Indicator. Blinking when collision occurs. Low for full duplex, and high for half duplex mode.
Link/Act	Link, Activity Indicator. Low for link established. Link/Act Blinking when the corresponding port is transmitting or receiving.
Spd1000	1000Mbps Speed Indicator. Low for 1000Mbps.
Spd100	100Mbps Speed Indicator. Low for 100Mbps.
Spd10	10Mbps Speed Indicator. Low for 10Mbps.
Spd1000/Act	1000Mbps Speed/Activity Indicator. Low for 1000Mbps. Blinking when the corresponding port is transmitting or receiving.
Spd100/Act	100Mbps Speed/Activity Indicator. Low for 100Mbps. Blinking when the corresponding port is transmitting or receiving.
Spd10/Act	10Mbps Speed/Activity Indicator. Low for 10Mbps. Blinking when the corresponding port is transmitting or receiving.
Spd100 (10)/Act	10/100Mbps Speed/Activity Indicator. Low for 10/100Mbps. Blinking when the corresponding port is transmitting or receiving.
Act	Activity Indicator. Act blinking when the corresponding port is transmitting or receiving.

The LED pin also supports pin strapping configuration functions. The PnLED0, PnLED1, and PnLED2 pins are dual-function pins: input operation for configuration upon reset, and output operation for LED after reset. If the pin input is pulled high upon reset, the pin output is active low after reset. If the pin input is pulled down upon reset, the pin output is active high after reset. For details refer to Figure 7, page 33, and Figure 8, page 33. Typical values for pull-up/pull-down resistors are 4.7K Ω .

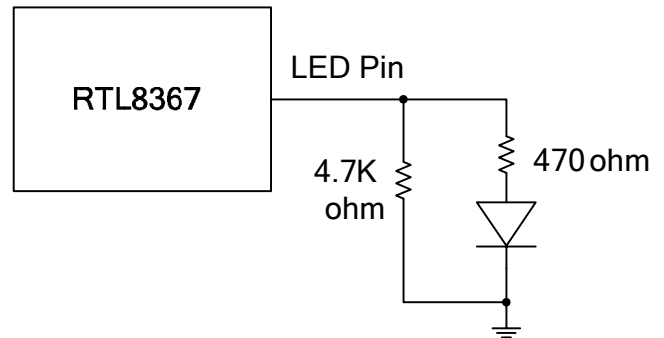
The PnLED1 can be combined with PnLED1 or PnLED2 as a Bi-color LED.

LED_ PnLED1 should operate with the same polarity as other Bi-color LED pins. For example:

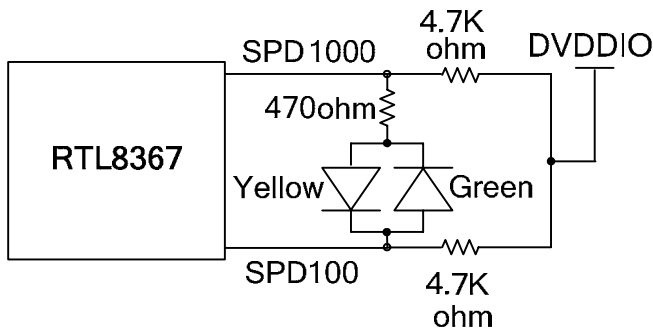
- P0LED1 should pull up upon reset if P0LED1 is combined with P0LED2 as a Bi-color LED, and P0LED2 input is pulled high upon reset. In this configuration, the output of these pins is active low after reset
- P0LED1 should be pulled down upon reset if P0LED1 is combined with P0LED2 as a Bi-color LED, and P0LED2 input is pulled down upon reset. In this configuration, the output of these pins is active high after reset

Pull-Up


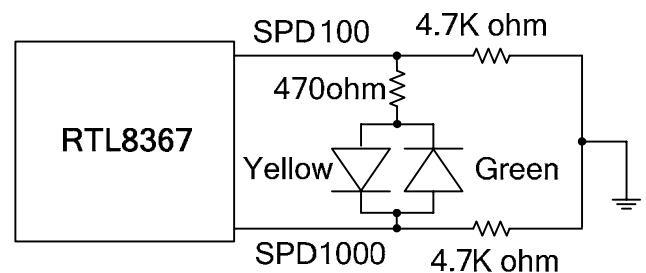
LED Pins Output Active Low

Pull-Down


LED Pins Output Active High

Figure 7. Pull-Up and Pull-Down of LED Pins for Single-Color LED
Pull-Up


LED Pins Output Active Low

Pull-Down


LED Pins Output Active High

Figure 8. Pull-Up and Pull-Down of LED Pins for Bi-Color LED

9.20. Green Ethernet

9.20.1. Link-On and Cable Length Power Saving

The RTL8367 provides link-on and dynamic detection of cable length and dynamic adjustment of power required for the detected cable length. This feature provides high performance with minimum power consumption.

9.20.2. Link-Down Power Saving

The RTL8367 implements link-down power saving on a per-port basis, greatly cutting power consumption when the network cable is disconnected. After it detects an incoming signal, it wakes up from link-down power saving and operates in normal mode.

9.21. IEEE 802.3az Energy Efficient Ethernet (EEE) Function

The RTL8367 supports IEEE 802.3az Energy Efficient Ethernet ability for 1000Base-T, 100Base-TX in full duplex operation, and 10Base-T in full/half duplex mode.

The Energy Efficient Ethernet (EEE) optional operational mode combines the IEEE 802.3 Media Access Control (MAC) sub-layer with 100Base-TX and 1000Base-T Physical Layers defined to support operation in Low Power Idle mode. When Low Power Idle mode is enabled, systems on both sides of the link can disable portions of the functionality and save power during periods of low link utilization.

- For 1000Base-T PHY: Supports Energy Efficient Ethernet with the optional function of Low Power Idle
- For 100Base-TX PHY: Supports Energy Efficient Ethernet with the optional function of Low Power Idle
- For 10Base-T, EEE defines a 10Mbps PHY (10Base-Te) with reduced transmit amplitude requirements. 10Base-Te is fully interoperable with 10Base-T PHYs over 100m of class-D (Cat-5) cable

The RTL8367 MAC uses Low Power Idle signaling to indicate to the PHY, and to the link partner, that a break in the data stream is expected, and components may use this information to enter power saving modes that require additional time to resume normal operation. Similarly, it informs the LPI Client that the link partner has sent such an indication.

10. Interface Descriptions

10.1. EEPROM SMI Host to EEPROM

The EEPROM interface of the RTL8367 uses the serial bus EEPROM Serial Management Interface (SMI) to read the EEPROM that space up to 256K-bits. When the RTL8367 is powered up, it drives SCK and SDA to read the registers from the EEPROM.

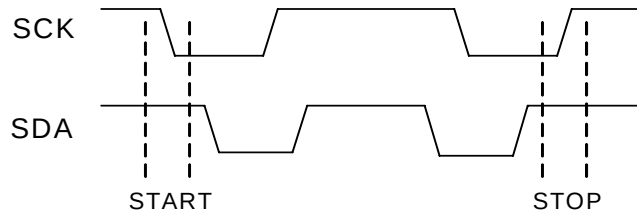


Figure 9. SMI Start and Stop Command

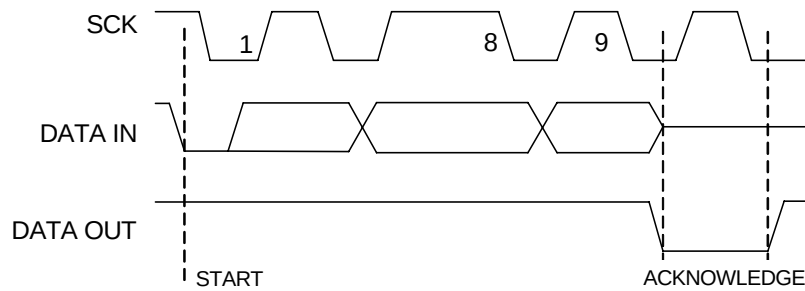


Figure 10. EEPROM SMI Host to EEPROM

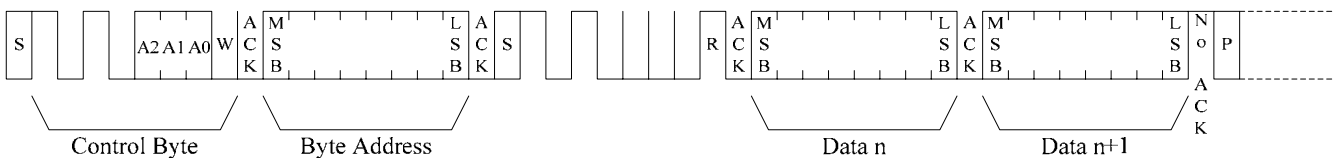


Figure 11. EEPROM SMI Host Mode Frame

10.2. EEPROM SMI Slave for External CPU

When EEPROM auto-load is complete, the RTL8367 registers can be accessed via SCK and SDA via an external CPU. The device address of the RTL8367 is 0x4. For the start and end of a write/read command, SCK needs one extra clock before/after the start/stop signals.

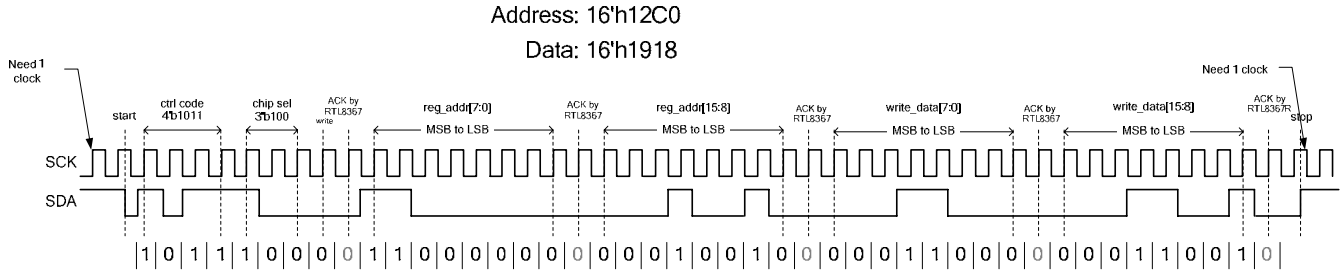


Figure 12. EEPROM SMI Write Command for Slave Mode

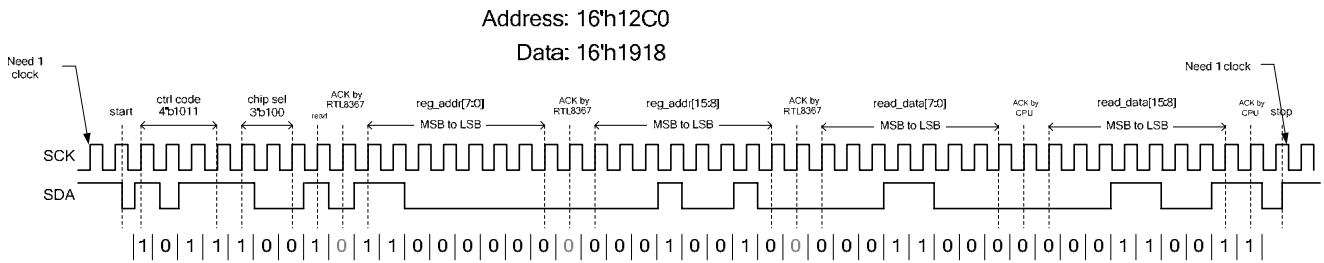


Figure 13. EEPROM SMI Read Command for Slave Mode

11. Register Descriptions

In this section the following abbreviations are used:

RO: Read Only

LH: Latch High until clear

RW: Read/Write

SC: Self Clearing

LL: Latch Low until clear

11.1. PCS Register (PHY 0~4)

Table 12. PCS Register (PHY 0~4)

Register	Register Description	Default
0	Control Register	0x1140
1	Status Register	0x7949
2	PHY Identifier 1	0x001C
3	PHY Identifier 2	0xC980
4	Auto-Negotiation Advertisement Register	0x0DE1
5	Auto-Negotiation Link Partner Ability Register	0x0000
6	Auto-Negotiation Expansion Register	0x0004
7	Auto-Negotiation Page Transmit Register	0x2001
8	Auto-Negotiation Link Partner Next Page Register	0x0000
9	1000Base-T Control Register	0x0E00
10	1000Base-T Status Register	0x0000
11~14	Reserved	0x0000
15	Extended Status	0x2000
16~31	ASIC Control Register	-

11.2. Register 0: Control

Table 13. Register 0: Control

Reg.bit	Name	Mode	Description	Default
0.15	Reset	RW/SC	1: PHY reset 0: Normal operation This bit is self-clearing.	0
0.14	Loopback (Digital loopback)	RW	1: Enable loopback. This will loopback TXD to RXD and ignore all activity on the cable media 0: Normal operation This function is usable only when this PHY is operated in 10Base-T full duplex, 100Base-TX full duplex, or 1000Base-T full duplex.	0
0.13	Speed Selection[0]	RW	[0.6, 0.13] Speed Selection[1:0] 11: Reserved 10: 1000Mbps 01: 100Mbps 00: 10Mbps This bit can be set through SMI (Read/Write).	0
0.12	Auto Negotiation Enable	RW	1: Enable auto-negotiation process 0: Disable auto-negotiation process This bit can be set through SMI (Read/Write).	1
0.11	Power Down	RW	1: Power down. All functions will be disabled except SMI function 0: Normal operation	0
0.10	Isolate	RW	1: Electrically isolates the PHY from GMII. The PHY is still able to respond to MDC/MDIO 0: Normal operation	0
0.9	Restart Auto Negotiation	RW/SC	1: Restart Auto-Negotiation process 0: Normal operation	0
0.8	Duplex Mode	RW	1: Full duplex operation 0: Half duplex operation This bit can be set through SMI (Read/Write).	1
0.7	Collision Test	RO	1: Collision test enabled 0: Normal operation When set, this bit will cause the COL signal to be asserted in response to the assertion of TXEN within 512-bit times. The COL signal will be de-asserted within 4-bit times in response to the de-assertion of TXEN.	0
0.6	Speed Selection[1]	RW	See bit 13	1
0.[5:0]	Reserved	RO	Reserved	000000

11.3. Register 1: Status

Table 14. Register 1: Status

Reg.bit	Name	Mode	Description	Default
1.15	100Base-T4	RO	0: No 100Base-T4 capability The RTL8367 does not support 100Base-T4 mode and this bit should always be 0.	0
1.14	100Base-TX-FD	RO	1: 100Base-TX full duplex capable 0: Not 100Base-TX full duplex capable	1
1.13	100Base-TX-HD	RO	1: 100Base-TX half duplex capable 0: Not 100Base-TX half duplex capable	1
1.12	10Base-T-FD	RO	1: 10Base-T full duplex capable 0: Not 10Base-TX full duplex capable	1
1.11	10Base-T-HD	RO	1: 10Base-T half duplex capable 0: Not 10Base-TX half duplex capable	1
1.10	100Base-T2-FD	RO	0: Not 100Base-T2 full duplex capable The RTL8367 does not support 100Base-T2 mode and this bit should always be 0.	0
1.9	100Base-T2-HD	RO	0: Not 100Base-T2 half duplex capable The RTL8367 does not support 100Base-T2 mode and this bit should always be 0.	0
1.8	Extended Status	RO	1: Extended status information in Register 15 The RTL8367 always supports Extended Status Register.	1
1.7	Reserved	RO	Reserved	0
1.6	MF Preamble Suppression	RO	The RTL8367 will accept management frames with preamble suppressed.	1
1.5	Auto-negotiate Complete	RO	1: Auto-negotiation process completed 0: Auto-negotiation process not completed	0
1.4	Remote Fault	RO/LH	1: Remote fault condition detected 0: No remote fault detected This bit will remain set until it is cleared by reading register 1 via the management interface.	0
1.3	Auto-Negotiation Ability	RO	1: Auto-negotiation capable (permanently =1)	1
1.2	Link Status	RO/LL	1: Link is established. If the link fails, this bit will be 0 until after reading this bit again 0: Link has failed since previous read If the link fails, this bit will be set to 0 until bit is read.	0
1.1	Jabber Detect	RO/LH	1: Jabber detected 0: No Jabber detected Jabber is supported only in 10Base-T mode.	0
1.0	Extended Capability	RO	1: Extended register capable (permanently =1)	1

11.4. Register 2: PHY Identifier 1

The PHY Identifier Registers #1 and #2 together form a unique identifier for the PHY section of this device. The Identifier consists of a concatenation of the Organizationally Unique Identifier (OUI), the vendor's model number, and the model revision number. A PHY may return a value of zero in each of the 32 bits of the PHY Identifier if desired. The PHY Identifier is intended to support network management.

Table 15. Register 2: PHY Identifier 1

Reg.bit	Name	Mode	Description	Default
2.[15:0]	OUI	RO	Composed of the 3 rd to 18 th bits of the Organizationally Unique Identifier (OUI), respectively.	0x001C

11.5. Register 3: PHY Identifier 2

Table 16. Register 3: PHY Identifier 2

Reg.bit	Name	Mode	Description	Default
3.[15:10]	OUI	RO	Assigned to the 19 th through 24 th bits of the OUI	110010
3.[9:4]	Model Number	RO	Manufacturer's model number	011000
3.[3:0]	Revision Number	RO	Manufacturer's revision number	0000

11.6. Register 4: Auto-Negotiation Advertisement

This register contains the advertisement abilities of this device as they will be transmitted to its Link Partner during Auto-negotiation.

Note: Each time the link ability of the RTL8367 is reconfigured, the auto-negotiation process should be executed to allow the configuration to take effect.

Table 17. Register 4: Auto-Negotiation Advertisement

Reg.bit	Name	Mode	Description	Default
4.15	Next Page	RO	1: Additional next pages exchange desired 0: No additional next pages exchange desired	0
4.14	Acknowledge	RO	Permanently=0	0
4.13	Remote Fault	RW	1: Advertises that the RTL8367 has detected a remote fault 0: No remote fault detected	0
4.12	Reserved	RO	Reserved	0
4.11	Reserved	RW	Reserved	0
4.10	Pause	RW	1: Advertises that the RTL8367 has flow control capability 0: No flow control capability	1
4.9	100Base-T4	RO	1: 100Base-T4 capable 0: Not 100Base-T4 capable (Permanently =0)	0
4.8	100Base-TX-FD	RW	1: 100Base-TX full duplex capable 0: Not 100Base-TX full duplex capable	1
4.7	100Base-TX	RW	1: 100Base-TX half duplex capable 0: Not 100Base-TX half duplex capable	1

Reg.bit	Name	Mode	Description	Default
4.6	10Base-T-FD	RW	1: 10Base-TX full duplex capable 0: Not 10Base-TX full duplex capable	1
4.5	10Base-T	RW	1: 10Base-TX half duplex capable 0: Not 10Base-TX half duplex capable	1
4.[4:0]	Selector Field	RO	[00001]=IEEE 802.3	00001

Note 1: The setting of Register 4 has no effect unless auto-negotiation is restarted or the link goes down.

Note 2: If 1000Base-T is advertised, then the required next pages are automatically transmitted.

11.7. Register 5: Auto-Negotiation Link Partner Ability

This register contains the advertised abilities of the Link Partner as received during Auto-negotiation. The content changes after a successful Auto-negotiation.

Table 18. Register 5: Auto-Negotiation Link Partner Ability

Reg.bit	Name	Mode	Description	Default
5.15	Next Page	RO	1: Link partner desires Next Page transfer 0: Link partner does not desire Next Page transfer	0
5.14	Acknowledge	RO	1: Link Partner acknowledges reception of Fast Link Pulse (FLP) words 0: Not acknowledged by Link Partner	0
5.13	Remote Fault	RO	1: Remote Fault indicated by Link Partner 0: No remote fault indicated by Link Partner	0
5.12	Reserved	RO	Reserved	0
5.11	Asymmetric Pause	RO	1: Asymmetric Flow control supported by Link Partner 0: No Asymmetric flow control supported by Link Partner. When auto-negotiation is enabled, this bit reflects Link Partner ability	0
5.10	Pause	RO	1: Flow control supported by Link Partner. 0: No flow control supported by Link Partner. When auto-negotiation is enabled, this bit reflects Link Partner ability	0
5.9	100Base-T4	RO	1: 100Base-T4 supported by Link Partner 0: 100Base-T4 not supported by Link Partner	0
5.8	100Base-TX-FD	RO	1: 100Base-TX full duplex supported by Link Partner 0: 100Base-TX full duplex not supported by Link Partner	0
5.7	100Base-TX	RO	1: 100Base-TX half duplex supported by Link Partner 0: 100Base-TX half duplex not supported by Link Partner	0
5.6	10Base-T-FD	RO	1: 10Base-TX full duplex supported by Link Partner 0: 10Base-TX full duplex not supported by Link Partner	0
5.5	10Base-T	RO	1: 10Base-TX half duplex supported by Link Partner 0: 10Base-TX half duplex not supported by Link Partner	0
5.[4:0]	Selector Field	RO	[00001]=IEEE 802.3	00000

11.8. Register 6: Auto-Negotiation Expansion

Table 19. Register 6: Auto-Negotiation Expansion

Reg.bit	Name	Mode	Description	Default
6.[15:5]	Reserved	RO	Ignore on read	0
6.4	Parallel Detection Fault	RO /LH	1: A fault has been detected via the Parallel Detection function 0: No fault has been detected via the Parallel Detection function	0
6.3	Link Partner Next Page Ability	RO	1: Link Partner is Next Page able 0: Link Partner is not Next Page able	0
6.2	Local Next Page Ability	RO	Not supported. Permanently =0	1
6.1	Page Received	RO /LH	1: A New Page has been received 0: A New Page has not been received	0
6.0	Link Partner Auto-Negotiation Ability	RO	If Auto-Negotiation is enabled, this bit means: 1: Link Partner is Auto-Negotiation able 0: Link Partner is not Auto-Negotiation able	0

11.9. Register 7: Auto-Negotiation Page Transmit Register

Table 20. Register 7: Auto-Negotiation Page Transmit Register

Reg.bit	Name	Mode	Description	Default
7.15	Next Page	RW	1: Link partner desires Next Page transfer 0: Link partner does not desire Next Page transfer	0
7.14	Reserved	RO	1: A fault has been detected via the Parallel Detection function 0: No fault has been detected via the Parallel Detection function	0
7.13	Message Page	RW	1: Message page 0: No Message page ability	1
7.12	Acknowledge 2	RW	1: Local device has the ability to comply with the message received 0: Local device has no ability to comply with the message received	0
7.11	Toggle	RO	Toggle bit	0
7.[10:0]	Message/ Unformatted Field	RW	Content of message/unformatted page	1

11.10. Register 8: Auto-Negotiation Link Partner Next Page Register

Table 21. Register 8: Auto-Negotiation Link Partner Next Page Register

Reg.bit	Name	Mode	Description	Default
8.15	Next Page	RO	Received Link Code Word Bit 15	0
8.14	Acknowledge	RO	Received Link Code Word Bit 14	0
8.13	Message Page	RO	Received Link Code Word Bit 13	0
8.12	Acknowledge 2	RO	Received Link Code Word Bit 12	0
8.11	Toggle	RO	Received Link Code Word Bit 11	0
8.[10:0]	Message/ Unformatted Field	RO	Received Link Code Word Bit 10:0	0

11.11. Register 9: 1000Base-T Control Register

Table 22. Register 9: 1000Base-T Control Register

Reg.bit	Name	Mode	Description	Default
9.[15:13]	Test Mode	RW	Test Mode Select. 000: Normal mode 001: Test mode 1 – Transmit waveform test 010: Test mode 2 – Transmit jitter test in MASTER mode 011: Test mode 3 – Transmit jitter test in SLAVE mode 100: Test mode 4 – Transmitter distortion test 101, 110, 111: Reserved	000
9.12	MASTER/SLAVE Manual Configuration Enable	RW	1: Enable MASTER/SLAVE manual configuration 0: Disable MASTER/SLAVE manual configuration	0
9.11	MASTER/SLAVE Configuration Value	RW	1: Configure PHY as MASTER during MASTER/SLAVE negotiation, only when bit 9.12 is set to logical one 0: Configure PHY as SLAVE during MASTER/SLAVE negotiation, only when bit 9.12 is set to logical one	1
9.10	Port Type	RW	1: Multi-port device 0: Single-port device	1
9.9	1000Base-T Full Duplex	RW	1: Advertise PHY is 1000Base-T full duplex capable 0: Advertise PHY is not 1000Base-T full duplex capable	1
9.8	1000Base-T Half Duplex	RW	1: Advertise PHY is 1000Base-T half duplex capable 0: Advertise PHY is not 1000Base-T half duplex capable	0
9.[7:0]	Reserved	RW	Reserved	0

11.12. Register 10: 1000Base-T Status Register

Table 23. Register 10: 1000Base-T Status Register

Reg.bit	Name	Mode	Description	Default
10.15	MASTER/SLAVE Configuration Fault	RO/LH/SC	1: MASTER/SLAVE configuration fault detected 0: No MASTER/SLAVE configuration fault detected	0
10.14	MASTER/SLAVE Configuration Resolution	RO	1: Local PHY configuration resolved to MASTER 0: Local PHY configuration resolved to SLAVE	0
10.13	Local Receiver Status	RO	1: Local receiver OK 0: Local receiver not OK	0
10.12	Remote Receiver Status	RO	1: Remote receiver OK 0: Remote receiver not OK	0
10.11	Link Partner 1000Base-T Full Duplex	RO	1: Link partner is capable of 1000Base-T full duplex 0: Link partner is not capable of 1000Base-T full duplex	0
10.10	1000Base-T Half Duplex	RO	1: Link partner is capable of 1000Base-T half duplex 0: Link partner is not capable of 1000Base-T half duplex	0
10.[9:8]	Reserved	RO	Reserved	0
10.[7:0]	Idle Error Count	RO/SC	Idle Error Counter. The counter stops automatically when it reaches 0xFF	0

11.13. Register 15: Extended Status

Table 24. Register 15: Extended Status

Reg.bit	Name	Mode	Description	Default
15.15	1000Base-X Full Duplex	RO	1: 1000Base-X full duplex capable 0: Not 1000Base-X full duplex capable	0
15.14	1000Base-X Half Duplex	RO	1: 1000Base-X half duplex capable 0: Not 1000Base-X half duplex capable	0
15.13	1000Base-T Full Duplex	RO	1: 1000Base-T full duplex capable 0: Not 1000Base-T full duplex capable	1
15.12	1000Base-T Half Duplex	RO	1: 1000Base-T half duplex capable 0: Not 1000Base-T half duplex capable	0
15.[11:0]	Reserved	RO	Reserved	0

12. Electrical Characteristics

12.1. Absolute Maximum Ratings

WARNING: Absolute maximum ratings are limits beyond which permanent damage may be caused to the device, or device reliability will be affected. All voltages are specified reference to GND unless otherwise specified.

Table 25. Absolute Maximum Ratings

Parameter	Min	Max	Units
Junction Temperature (Tj)	-	+125	°C
Storage Temperature	-45	+125	°C
DVDDIO and AVDDH Supply Referenced to GND and AGND	GND-0.3	+3.63	V
DVDDL, AVDDL, PLLVDDL0, PLLVDDL1 Supply Referenced to GND, AGND, PLLGND0, and PLLGND1	GND-0.3	+1.1	V
Digital Input Voltage	GND-0.3	VDDIO+0.3	V

12.2. Recommended Operating Range

Table 26. Recommended Operating Range

Parameter	Min	Typical	Max	Units
Ambient Operating Temperature (Ta)	0	-	70	°C
DVDDIO and AVDDH Supply Voltage Range	3.135	3.3	3.465	V
DVDDL, AVDDL, PLLVDDL0, PLLVDDL1 Supply Voltage Range	0.95	1.0	1.05	V

12.3. Thermal Characteristics

12.3.1. QFP-128

12.3.1.1 Assembly Description

Table 27. Assembly Description

Package	Type	QFP128
	Dimension (L x W)	14x20mm
	Thickness	1.4mm
PCB	PCB Dimension (L x W)	95 x 75mm
	PCB Thickness	1.6mm
	Number of Cu Layer-PCB	2-Layer: - Top layer (1oz): 20% coverage of Cu - Bottom layer (1oz): 75% coverage of Cu 4-Layer: - 1st layer (1oz): 20% coverage of Cu - 2nd layer (1oz): 80% coverage of Cu - 3rd layer (1oz): 80% coverage of Cu - 4th layer (1oz): 75% coverage of Cu

12.3.1.2 Material Properties

Table 28. Material Properties

Item		Material	Thermal Conductivity K (W/m-k)
Package	Die	Si	147
	Silver Paste	1033BF	2.5
	Lead Frame	CDA7025	168
	Mold Compound	7372	0.88
PCB		Cu	400
		FR4	0.2

12.3.1.3 Simulation Conditions

Table 29. Simulation Conditions

Input Power	1.8W
Test Board (PCB)	2L (2S)/4L (2S2P)
Control Condition	Air Flow = 0, 1, 2, m/s

12.3.1.4 Thermal Performance of QFP-128 on PCB under Still Air Convection

Table 30. Thermal Performance of QFP-128 on PCB under Still Air Convection

	θ_{JA}	θ_{JB}	θ_{JC}	Ψ_{JT}	Ψ_{JB}
4L PCB	34.9	24.5	14.2	4.5	23.1
2L PCB	39.1	25.0	14.6	6.7	23.4

Note:

θ_{JA} : Junction to ambient thermal resistance

θ_{JB} : Junction to board thermal resistance

θ_{JC} : Junction to case thermal resistance

Ψ_{JT} : Junction to top center of package thermal characterization

Ψ_{JB} : Junction to bottom surface center of PCB thermal characterization

12.3.1.5 Thermal Performance of QFP-128 on PCB under Forced Convection

Table 31. Thermal Performance of QFP-128 on PCB under Forced Convection

	Air Flow (m/s)	0	1	2
4L PCB	θ_{JA}	34.9	31	29.5
	Ψ_{JT}	4.5	5.2	5.4
	Ψ_{JB}	23.1	22.5	22.3
2L PCB	θ_{JA}	39.1	34	31.9
	Ψ_{JT}	6.7	7.7	8.1
	Ψ_{JB}	23.4	22.8	22.4

12.4. DC Characteristics

Table 32. DC Characteristics

Parameter	SYM	Min	Typical	Max	Units
System Idle (All UTP Port Link Down, without Extension ports and LEDs)					
Power Supply Current for VDDH	I_{DVDDIO}, I_{AVDDH}	-	-	TBD	mA
Power Supply Current for VDDL	$I_{DVDDL}, I_{AVDDL}, I_{PLLVDL}$	-	-	TBD	mA
1000M Active (All UTP Ports Link/Active, without Extension ports and LEDs)					
Power Supply Current for VDDH	I_{DVDDIO}, I_{AVDDH}	-	-	350	mA
Power Supply Current for VDDL	$I_{DVDDL}, I_{AVDDL}, I_{PLLVDL}$	-	-	900	mA
VDDIO=3.3V					
TTL Input High Voltage	V_{ih}	1.9	-	-	V
TTL Input Low Voltage	V_{il}	-	-	0.7	V
Output High Voltage	V_{oh}	2.7	-	-	V
Output Low Voltage	V_{ol}	-	-	0.6	V
VDDIO=2.5V					
TTL Input High Voltage	V_{ih}	1.7	-	-	V
TTL Input Low Voltage	V_{il}	-	-	0.7	V
Output High Voltage	V_{oh}	2.25	-	-	V
Output Low Voltage	V_{ol}	-	-	0.4	V

Note: $DVDDIO=AVDDH=3.3V$, $DVDDL=AVDDL=PLLVDL0=PLLVDL1=1.0V$.

12.5. AC Characteristics

12.5.1. EEPROM SMI Host Mode Timing Characteristics

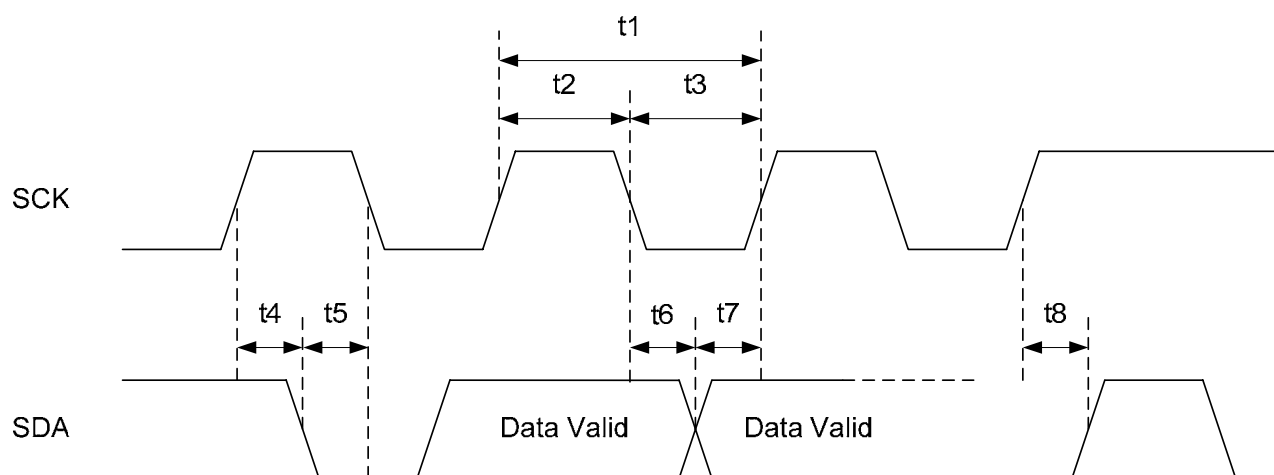


Figure 14. EEPROM SMI Host Mode Timing Characteristics

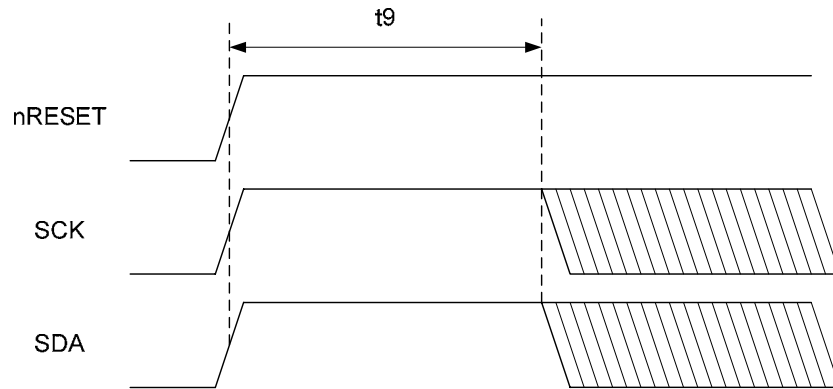


Figure 15. SCK/SDA Power on Timing

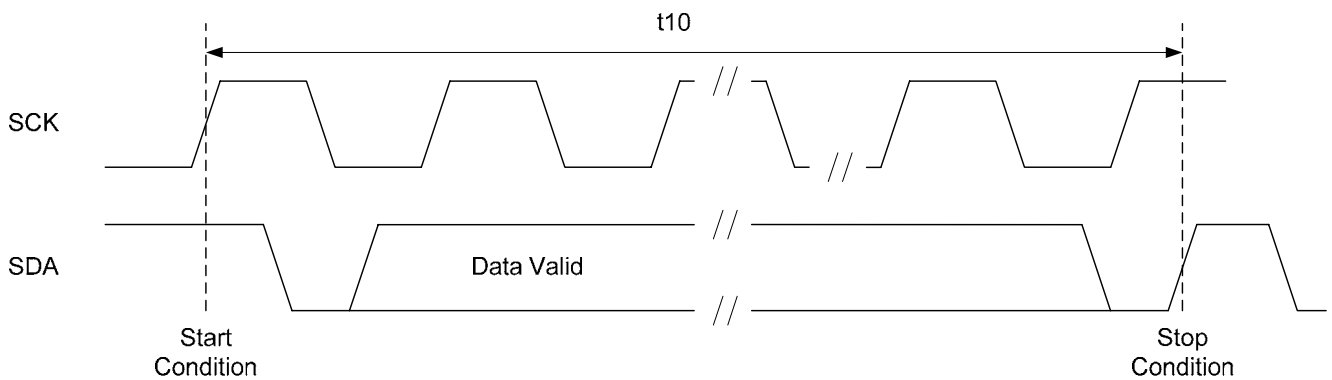


Figure 16. EEPROM Auto-Load Timing

Table 33. EEPROM SMI Host Mode Timing Characteristics

Symbol	Description	Type	Min	Typical	Max	Units
t1	SCK Clock Period	O	TBD	TBD	-	μs
t2	SCK High Time	O	TBD	TBD	-	μs
t3	SCK Low Time	O	TBD	TBD	-	μs
t4	START Condition Setup Time	O	TBD	TBD	-	μs
t5	START Condition Hold Time	O	TBD	TBD	-	μs
t6	Data Hold Time	O	-	TBD	-	μs
t7	Data Setup Time	O	-	TBD	-	μs
t8	STOP Condition Setup Time	O	TBD	TBD	-	μs
t9	SCK/SDA Active from Reset Ready	O	TBD	TBD	-	ms
t10	8K-bits EEPROM Auto-Load Time	O	TBD	TBD	-	ms
-	SCK Rise Time (10% to 90%)	O	-	TBD	TBD	ns
-	SCK Fall Time (90% to 10%)	O	-	TBD	TBD	ns
-	Duty Cycle	O	TBD	TBD	TBD	%

Note: t6, t7, and t10 are measured with ATMEL AT24C08 EEPROM.

12.5.2. EEPROM SMI Slave Mode Timing Characteristics

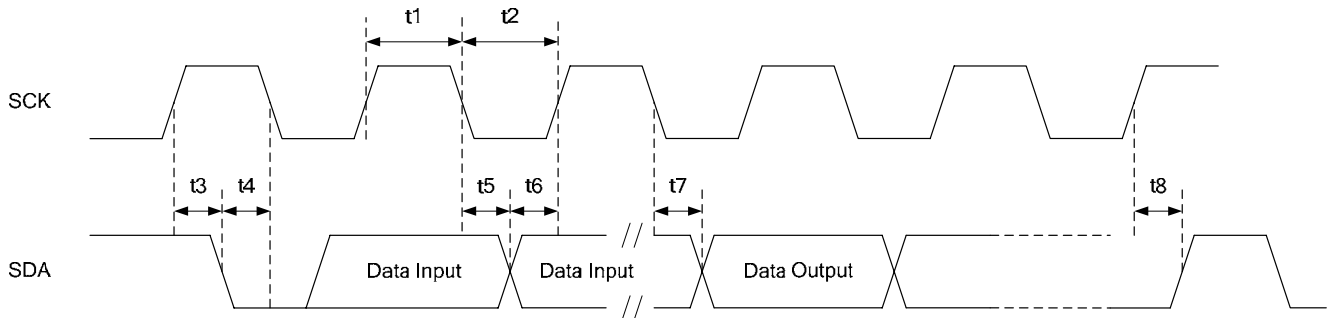


Figure 17. EEPROM SMI Slave Mode Timing Characteristics

Table 34. EEPROM SMI Slave Mode Timing Characteristics

Symbol	Description	Type	Min	Typical	Max	Units
t1	SCK High Time	I	TBD	-	-	μ s
t2	SCK Low Time	I	TBD	-	-	μ s
t3	START Condition Setup Time	I	TBD	-	-	μ s
t4	START Condition Hold Time	I	TBD	-	-	μ s
t5	Data Hold Time	I	TBD	-	-	μ s
t6	Data Setup Time	I	TBD	-	-	ns
t7	Clock to Data Output Delay	O	-	TBD	-	ns
t8	STOP Condition Setup Time	I	TBD	-	-	μ s

12.5.3. MDIO Slave Mode Timing Characteristics

The RTL8367 supports MDIO slave mode. The Master (the RTL8367 link partner CPU) can access the Slave (RTL8367) registers via the MDIO interface. The MDIO is a bi-directional signal that can be sourced by the Master or the Slave. In a write command, the Master sources the MDIO signal. In a read command, the Slave sources the MDIO signal.

- The timing characteristics (t1, t2, and t3 in Table 35) of the Master (the RTL8367 link partner CPU) are provided by the Master when the Master sources the MDIO signal (Write command)
- The timing characteristics (t4 in Table 35) of the Slave (RTL8367) are provided by the RTL8367 when the RTL8367 sources the MDIO signal (Read command)

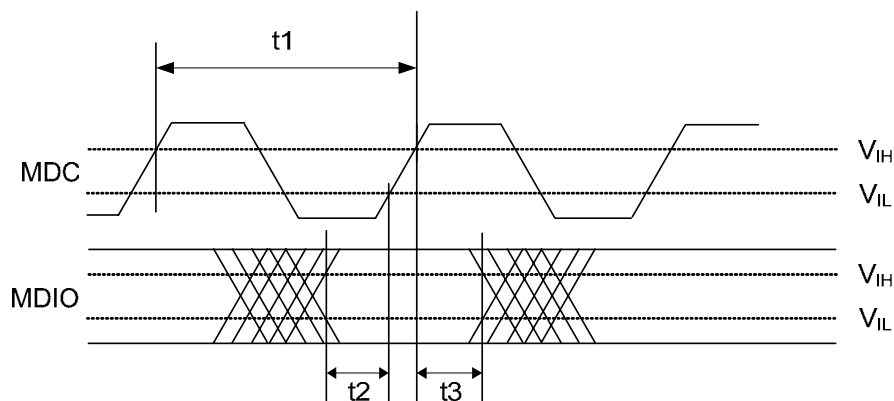


Figure 18. MDIO Sourced by Master (RTL8367 Link Partner CPU)

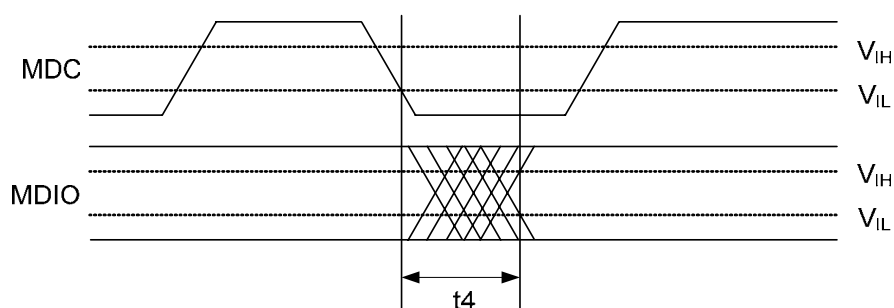


Figure 19. MDIO Sourced by Slave (RTL8367)

Table 35. MDIO Timing Characteristics and Requirements

Parameter	SYM	Description/Condition	Type	Min	Typical	Max	Units
MDC Clock Period	t1	Clock Period	I	125	-	-	ns
MDIO to MDC Rising Setup Time (Write Data)	t2	Input Setup Time	I	26	-	-	ns
MDIO to MDC Rising Hold Time (Write Data)	t3	Input Hold Time	I	26	-	-	ns
MDC to MDIO Delay Time (Read Data)	t4	Clock (Falling Edge) to Data Delay Time	O	0	-	40	ns

12.6. Power and Reset Characteristics

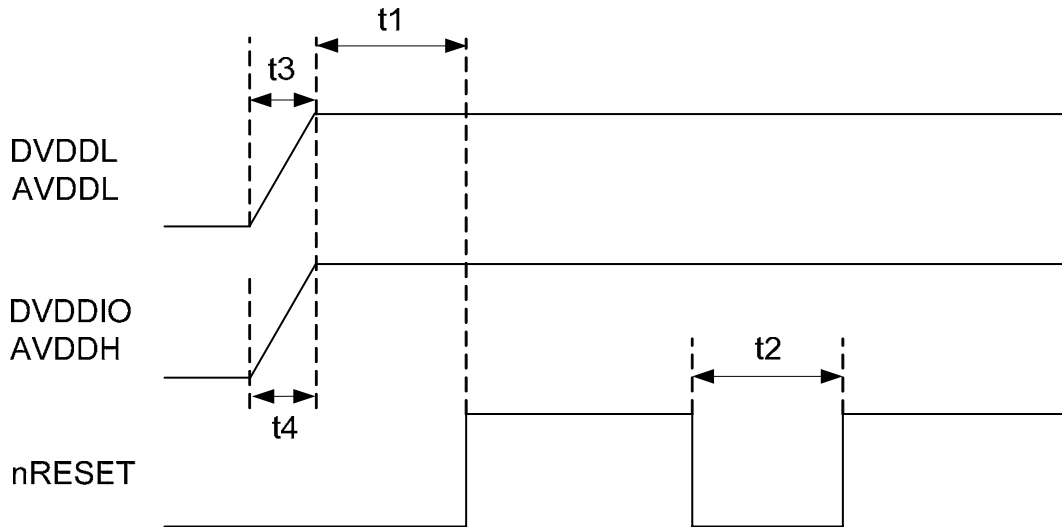


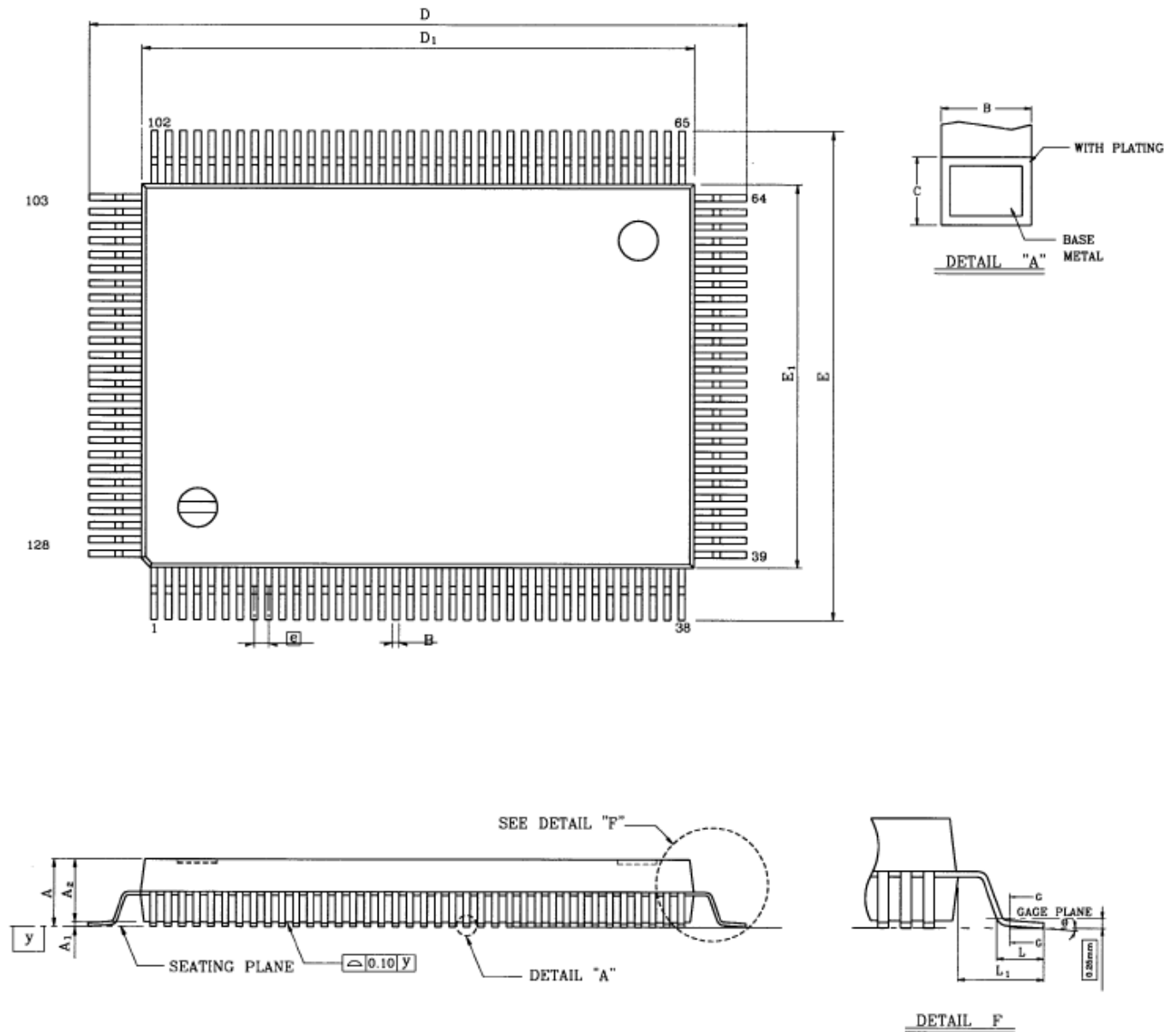
Figure 20. Power and Reset Characteristics

Table 36. Power and Reset Characteristics

Parameter	SYM	Description/Condition	Type	Min	Typical	Max	Units
Reset Delay Time	t1	The duration from all powers steady to the reset signal released to high.	I	10	-	-	ms
Reset Low Time	t2	The duration of reset signal remain low time for issuing a reset to RTL8367.	I	10	-	-	ms
VDDL Power Rising Settling Time	t3	DVDDL and AVDDL power rising settling time.	I	1	-	-	ms
VDDH Power Rising Settling Time	t4	DVDDIO and AVDDH power rising settling time.	I	1	-	-	ms

13. Mechanical Dimensions

Thermally Enhanced Quad Flat Package (QFP) 128 Leads 14x20mm Outline.



13.1. Mechanical Dimensions Notes

Symbol	Dimension in mm			Dimension in inch		
	Min	Nom	Max	Min	Nom	Max
A	-	-	3.40	-	-	0.134
A ₁	0.25	-	-	0.010	-	-
A ₂	2.73	2.85	2.97	0.107	0.112	0.117
B	0.17	0.22	0.27	0.007	0.009	0.011
C	0.09	-	0.20	0.004	-	0.008
D	23.00	23.20	23.40	0.906	0.913	0.921
D ₁	19.90	20.00	20.10	0.783	0.787	0.791
E	17.00	17.20	17.40	0.669	0.677	0.685
E ₁	13.90	14.00	14.10	0.547	0.551	0.555
e	0.50BSC			0.020BSC		
L	0.73	0.88	1.03	0.029	0.035	0.041
L1	1.60BSC			0.063BSC		
y	-	-	0.10	-	-	0.004
θ	0	-	7°	0	-	7°

Note 1: CONTROLLING DIMENSION: MILLIMETER (mm).

Note 2: REFERENCE DOCUMENT: JEDEC MS-26.

14. Ordering Information

Table 37. Ordering Information

Part Number	Package	Status
RTL8367-VB-CG	QFP 128-Pin 'Green' Package	-

Note: See page 7 for package identification.

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